
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) remains one of the most widely adopted neighbor-embedding procedures, in large part because its optimization can be interpreted through the coupled action of attractive and repulsive forces among high-dimensional observations as they are relocated into a low-dimensional representation. In this paper, we analyze these forces as explicit functional objects, thereby clarifying how they shape cluster formation, visualization geometry, and the behavior of UMAP relative to contemporary embedding methods. Repulsion primarily accentuates dissimilarity, delimiting cluster boundaries and modulating inter-cluster separation, whereas attraction exhibits a more intricate role because attractive tension between neighboring points can, after discretized optimization, manifest simultaneously as contraction and expansion in the embedded space. This observation accounts for the practical importance of learning-rate annealing and motivates asymmetric treatment of attractive and repulsive terms. Moreover, by modifying the attraction profile, we improve the consistency of cluster formation under random initialization. Overall, the analysis renders UMAP and related embedding methods more interpretable, robust, and accurate.

1 Introduction

The contemporary data landscape is marked by the routine production of observations whose dimensionality far exceeds the capacity of direct visual inspection. Dimensionality reduction (DR) techniques have consequently become indispensable instruments for exploratory analysis, because they translate latent high-dimensional organization into visually accessible low-dimensional structure. Among these methods, t -distributed stochastic neighbor embedding (Maaten and Hinton 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al. 2018) are especially prominent, and both are governed by attraction–repulsion dynamics that draw similar observations together while displacing dissimilar ones. Because these algorithms are unsupervised, they do not depend on labels; rather, they infer and preserve intrinsic high-dimensional structure by exploiting local attractive relationships and global repulsive relationships. This property makes them particularly suitable for clustering (Becht et al. 2019), exploratory data analysis (Fleischer and Islam 2020), anomaly detection in semiconductor manufacturing (Fan et al. 2021), visual search (González-Márquez et al. 2024), time series analysis (Altin and Cakir 2024), the study of representation convergence (Huh et al. 2024), and outlier image detection (Islam and Fleischer 2024), where the visualization of hidden regularities in unlabeled data is both scientifically meaningful and practically consequential. By learning embeddings in a data-driven, label-free manner, DR exemplifies how unsupervised methods can condense complex datasets into interpretable geometric summaries.

Building on this attraction–repulsion principle, a succession of newer methods has appeared (Amid and Warmuth 2019; Agrawal et al. 2021; Wang et al. 2021; Narayan et al. 2021; Yang et al. 2022; Wang et al. 2025; Kury et al. 2025), with each method emphasizing different geometric or statistical aspects of the data. Despite their broad utility, however, these approaches often depend on heuristic design choices whose visual consequences are difficult to interpret unambiguously. Moreover, because DR necessarily introduces distortions (Chari and Pachter 2023), a more precise understanding of the algorithms is essential if practitioners are to interpret embeddings responsibly, avoid spurious structures, and tune performance effectively. In practice, compact clusters are obtained through a mixture of initialization strategies, learning-rate schedules, and kernel-function parameters, yet the force-level dynamics that generate those clusters have not been systematically characterized. The difficulty is compounded by the fact that central tunable quantities are embedded in abstract functional forms, which obscures their mechanistic interpretation.

In this paper, we decompose the embedding forces into their constituent components and extract the functional shapes governing attraction and repulsion. This decomposition reveals that the need for learning-rate annealing, the difficulty of obtaining consistent outputs from random initialization, and the emergence of cluster structure are all principally mediated by attraction, whereas repulsion chiefly controls inter-cluster separation. Our specific contributions are as follows:

1. We derive attraction and repulsion shapes from the corresponding forces, establish conditions under which distances contract or expand, and thereby provide a force-theoretic perspective on neighbor-embedding algorithms (Section 4).
2. We demonstrate that UMAP’s attraction shape induces the counterintuitive coexistence of contraction and expansion in pairwise distances, and, by comparing attraction shapes across algorithms, we clarify how attraction governs learning-rate annealing (Section 5).
3. We modify the attraction shape to improve embedding consistency under random initialization, thereby indicating that UMAP encodes a reproducible underlying structure (Section 5.1).
4. We analyze repulsion shapes to obtain a more detailed account of cluster compactness and the regulation of inter-cluster distance (Section 5.2).
5. We compare the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG- t -SNE, identify their commonalities and distinctions, and characterize the stability consequences of these differences (Section 6).

The main text concentrates on UMAP, which has become a de facto standard across many fields, while incorporating other algorithms where they illuminate the force-shape analysis. Additional algorithmic details are provided in the appendix, and the code used for the study will be released upon acceptance.

2 Related Works

The lineage of modern iterative, graph-based neighbor-embedding algorithms can be traced to stochastic neighbor embedding (SNE) (Hinton and Roweis, 2002) and its t -distributed extension (t -SNE) (Maaten and Hinton, 2008). Both methods construct dense graphs in which every data point maintains a pairwise relationship with every other point, irrespective of whether the two observations are genuinely similar, and the resulting graph weights are normalized so that they can be interpreted probabilistically. In parallel, methods such as locally linear embedding (Roweis and Saul, 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi, 2002) relied instead on k -nearest neighbor (k -NN) graphs of pairwise interactions; as spectral methods, these algorithms obtain embeddings through eigenvalue decomposition. Subsequent work by Tang et al. (Tang et al., 2016) incorporated the k -NN graph into an iterative optimization procedure while removing normalization in the low-dimensional space. UMAP, introduced by McInnes et al. (2018), extended this development by removing normalization in both the high- and low-dimensional spaces and by obtaining the embedding through pairwise interactions alone. Its optimization uses an explicit attractive force to preserve local neighborhoods and a repulsive force to separate dissimilar observations, and later methods, including PaCMAP (Wang et al., 2021) and NEG- t -SNE (Damrich et al., 2023), build on this foundation. For a recent survey of methods, see de Bodt et al. (2025).

Substantial theoretical work has sought to clarify the relationships among these algorithms. An early analysis of SNE established that, when data are well clustered in the original space, the embedding also preserves clustering structure (Shaham and Steinerberger, 2017). A related analysis of t -SNE by Linderman and Steinerberger (2019) showed that the number of clusters visible in the embedding is a lower bound on the number of clusters present in the original space, a result subsequently refined through additional characterizations (Arora et al., 2018; Cai and Ma, 2022; Linderman and Steinerberger, 2022). Because t -SNE and UMAP arise from a shared conceptual framework yet often yield markedly different visualizations, a central theme in the literature has been the identification of their common structure (Böhm et al., 2022; Damrich et al., 2023; Draganov et al., 2023). Böhm et al. (2022) proposed that Laplacian Eigenmaps, ForceAtlas2 (Jacomy et al., 2014), UMAP, and t -SNE can be understood as samples from a common methodological spectrum. Consistent with this view, LE and t -SNE are connected through the early-exaggeration phase (Cai and Ma, 2022), while the relationship between UMAP and t -SNE can be expressed through contrastive estimation (Damrich et al., 2023). Around the same time, Hu et al. (2023) independently identified a connection between contrastive learning and SNE. More recent explanations of dimensionality reduction have introduced probabilistic perspectives (Ravuri et al., 2023; Ravuri and Lawrence, 2024), kernel-based analyses (Draganov and Dohn, 2023), and information-geometric interpretations (Kolpakov and Rocke, 2024).

3 Uniform Manifold Approximation and Projection

UMAP constructs a high-dimensional graph for the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ by assigning pairwise weights according to the relation $p_{i,j} = f_h(d(x_i, x_j))$, which typically lies in $[0, 1]$ (for details, see Appendix H).

The corresponding graph in the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is specified through the differentiable affinity function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b determine the density of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d regulates the distance between the two nearest low-dimensional points.

UMAP aims to minimize the following cross-entropy loss function:

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term induces attraction, whereas the second induces repulsion. Rather than optimizing all point pairs at every iteration, UMAP adopts negative sampling (Mikolov et al. 2013; Tang et al. 2016): for each edge with $p_{ij} > 0$, which we refer to as a positive edge, the algorithm randomly samples several negative edges. The attractive update is applied to the positive edge,

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

whereas the repulsive update is applied to the sampled negative edges,

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t is the step number. Note that y_j is not updated for negative edges. For a detailed analysis of UMAP's loss, see Damrich and Hamprecht (2021).

4 Attraction and Repulsion Shapes

The action of the updates (4-6) can be simplified by decomposing the gradients $\nabla_{y_i^t} \log(q_{ij})$ ($\nabla_{y_i^t} \log(1 - q_{ij})$) into a distance-dependent scalar coefficient multiplying the displacement vector $(y_i - y_j)$. We refer to this scalar coefficient as the attraction (repulsion) shape. Although UMAP provides the principal example, the same formalism applies to any embedding method whose optimization is organized around attraction and repulsion.

By writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, where $\zeta = \|y_i - y_j\|_2$, we can update the equations of a positive edge as

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ denotes the attraction shape, and we use the Euclidean identity $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. Analogously, the update equation for a negative edge can be written as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Related decompositions have appeared previously, for example in (McInnes et al. 2018; Agrawal et al. 2021; Draganov and Dohn 2023), although their mathematical presentation and interpretive use differ. The original UMAP paper (McInnes et al. 2018) employed the decomposition primarily as a computational device for efficient processing, whereas Draganov and Dohn (2023) used it to compare algorithms through their kernels; in both cases, the derivative itself was the central object. By contrast, we regard the decomposed coefficients as independent functional shapes that may assume distinct forms. A related strategy was adopted by Agrawal et al. (2021), who expressed the force as a scalar coefficient times the unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ in order to emphasize magnitude. The analysis below shows that the shape-level decomposition is more informative than force magnitude alone.

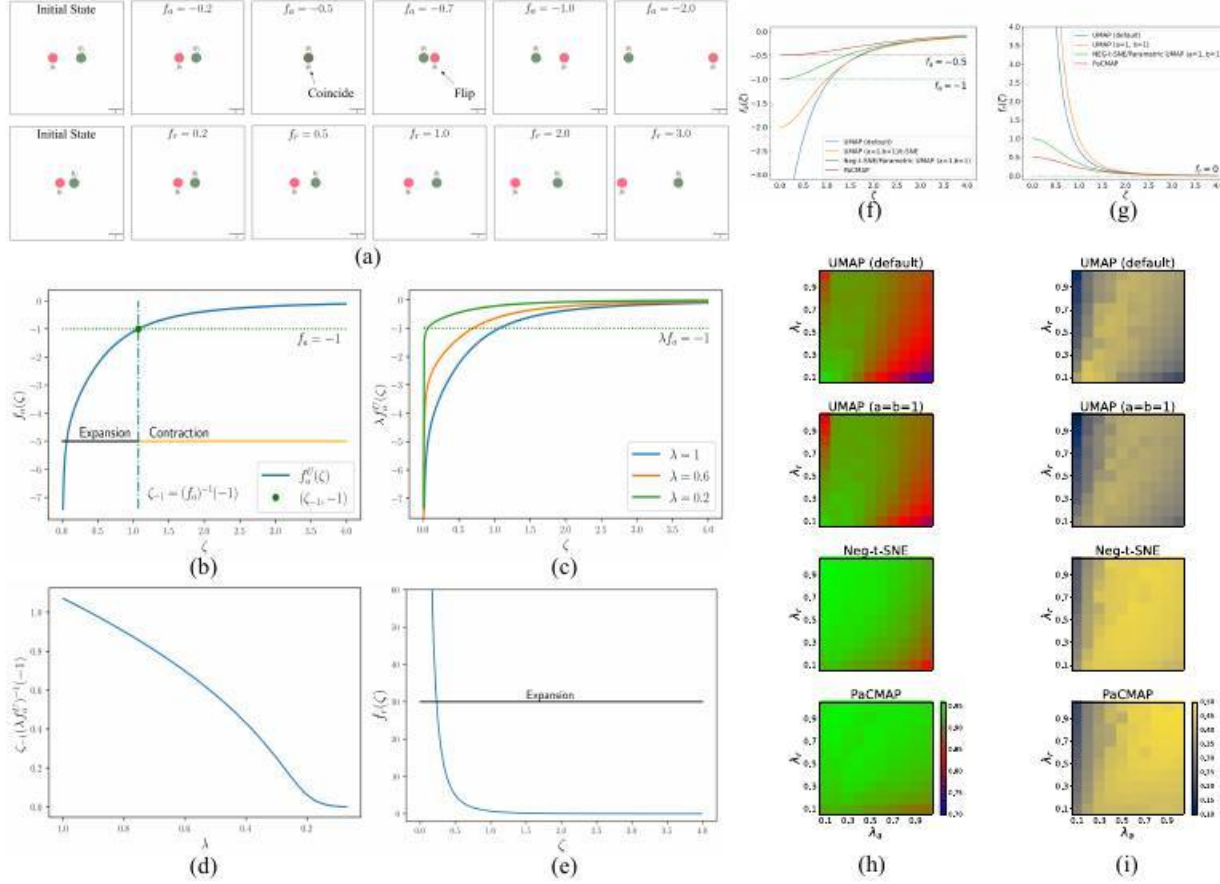


Figure 1: Attraction and repulsion shapes in UMAP. (a) Effect of different values of f_a (top) and f_r (bottom) on a pair. (b) Attraction shape of UMAP. (c) Effective attraction shape (λf_a) for various learning rates λ . (d) Minimum distance for contraction (ζ_{-1}) as λ decreases. (e) Repulsion shape of UMAP. (f) Attraction and (g) repulsion shapes of various embedding methods. (h) Trustworthiness and (i) Silhouette Score of various methods as the constant learning rate is varied for attraction (λ_a) and repulsion (λ_r) independently. Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

4.1 Conditions for Attraction and Repulsion

We next derive conditions for attraction and repulsion directly from the update equations (7)-(10). The first proposition characterizes when the distance between y_i^t and y_j^t on a positive edge contracts:

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Thus, λf_a functions as the effective attraction shape, although in most settings the unscaled shape f_a is already sufficient for interpreting embedding behavior. For a negative edge, the corresponding proposition characterizes expansion of the distance between y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Including a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ in Eq. (10) would not change the conclusion of this proposition. These conditions should be interpreted as per-iteration certificates for guaranteed contraction or expansion, rather than as statements about learning-rate tuning or decay; indeed, one can design shapes for which contraction and expansion do not depend on learning-rate annealing.

Figure 1(a) illustrates how different values of $f_a < 0$ and $f_r > 0$ affect a pair of points. The repulsion shape is comparatively transparent because any positive value increases the pairwise distance. The attraction shape is subtler: f_a encodes both contraction and expansion, so that distances contract for $f_a \in (-1.0, 0)$, with the sign of the point ordering

flipping at the maximally attractive value $f_a = -0.5$ when the points coincide. Values below -1.0 instead increase the distance. Although these forces are applied locally, their accumulated action determines the global geometry of the embedding.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Combining the gradient decomposition with the distance-based affinity in (1) yields the following attraction and repulsion shapes:

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section examines the default UMAP shapes, which are governed primarily by the parameters a and b , and then uses perturbations of those parameters to expose their geometric consequences. Unless otherwise noted, the discussion applies when $b \leq 1$, for which the attraction shape is strictly increasing and therefore invertible (for derivation, see Appendix A).

Figure 1(b) displays the default UMAP attraction shape ($a = 1.58, b = 0.89$), which becomes unbounded as $\zeta \rightarrow 0$. Consistent with Proposition 4.1 the transition from contraction to expansion occurs when λf_a^U crosses -1 in the small-distance regime. Because the attraction shape is invertible, the transition distance $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$ can be interpreted as the minimum distance at which attractive updates still contract the pair. Thus, $\zeta > \zeta_{-1}$ produces contraction, whereas $0 < \zeta < \zeta_{-1}$ produces expansion, contradicting the naive expectation that attractive updates invariably bring neighboring points closer together.

When ζ_{-1} is large, neighboring points can oscillate between contraction and expansion, producing visually diffuse or fuzzy clusters. Consequently, the optimization objective can be reframed, for sharp-boundary embeddings, as the pursuit of the limiting regime $\zeta_{-1} \rightarrow 0$. This reframing explains why the UMAP learning-rate schedule anneals toward zero (Figs. 1(c,d)). By contrast, the repulsion shape (Eq. 12) is always positive and therefore satisfies Proposition 4.2: f_r^U approaches 0 as $\zeta \rightarrow \infty$ and diverges as $\zeta \rightarrow 0$ (Fig. 1(e)).

Comparing attraction shapes across algorithms shows that UMAP and t-SNE are the primary methods confronted with the regime $f_a < -1$ (and consequently $\zeta_1 > 0$ at $\lambda = 1$, Fig. 1(f)). t-SNE mitigates this issue by weighting updates with the corresponding p_{ij} values, whereas UMAP relies on learning-rate annealing. In contrast, methods such as Neg-t-SNE and PaCMAP place f_a naturally within $[-1, 0]$ and therefore satisfy Proposition 4.1 for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$; PaCMAP’s weighted f_a , which remains below 0.5, further prevents flips during attraction. The repulsion-shape comparison also shows that UMAP is distinctive in producing very large values at small distances (Fig. 1(g)); although the shape is unbounded, these values are often clipped during optimization. Additional details for these and related methods are provided in Appendix G.

We evaluate the consequences of these attraction–repulsion choices on MNIST by assigning separate constant learning rates to attraction (λ_a) and repulsion (λ_r) and varying them independently (Figs. 1(h,i)). For UMAP, smaller attraction learning rates (< 0.5 , and consequently $\zeta_{-1} \simeq 0$) yield better embeddings, as measured by trustworthiness (T) (Venna and Kaski, 2001), a measure of structure preservation (Fig. 1(h)), and better clustering, as measured by silhouette score (SIL) (Rousseeuw, 1987), a measure of cluster separation (Fig. 1(i)); by contrast, essentially the full range of $\lambda_r \in [0, 1]$ remains usable. For NEG-t-SNE and PaCMAP, where $\zeta_{-1} = 0$, the full parameter range is effective. These results confirm that UMAP depends on driving ζ_{-1} close to zero for satisfactory embedding and clustering, which it achieves in practice through learning-rate annealing, whereas $\zeta_{-1} = 0$ is already built into newer algorithms. Additional discussion, results, and dataset-specific illustrations appear in Section 6 and Appendices B, G, and I.

5.1 Improving UMAP’s Consistency under Random Initialization

The consistency of UMAP embeddings is strongly conditioned on initialization (Kobak and Linderman, 2021; Wang et al., 2021). In standard practice, the embedding is initialized either by principal component analysis (PCA) of the data or by spectral decomposition of the high-dimensional graph, choices that produce relatively stable mappings despite the stochasticity of the optimization. Under random initialization, by contrast, clusters may fail to form or may appear in a different orientation at each execution. When two high-dimensional nearest neighbors are initially far apart in the low-dimensional space, their attractive force can be too weak to contract them effectively. This near-sightedness (Wang

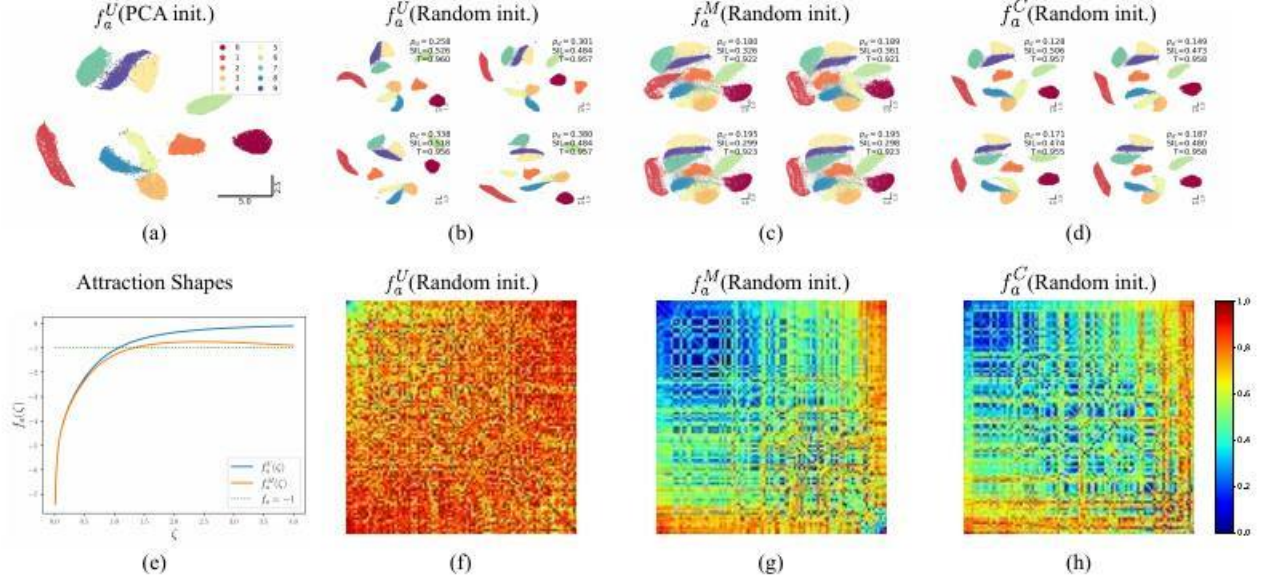


Figure 2: Effect of random initialization under different attraction shapes for the MNIST dataset. (a) PCA-initialized mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) UMAP (0.78 ± 0.13), (g) modified (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20) shapes. Diagonal (i, i) entries are sorted by p_d from (a), whereas off-diagonal (i, j) entries report p_d between the i^{th} and j^{th} mappings. The matrices and (mean $p_d \pm \text{std}$) summaries indicate that default UMAP embeddings are not self-similar, while modified and composite attraction shapes promote scale-invariant structure.

et al. [2021] is visible directly in the attraction shape, because $|f_a^U|$ diminishes toward zero as distance increases (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)| \zeta = 0$).

A natural remedy is to induce “far-sightedness” by strengthening attraction at large distances, thereby enabling distant high-dimensional neighbors to approach one another in the embedding. To test this hypothesis, we modify the UMAP attraction shape by increasing the attractive force:

$$f_a^M = f_a^U - \beta \zeta, \quad (13)$$

where β regulates the strength of the added term (we used $\beta = 0.2$, Fig. 2(e)). This alteration of the attraction shape is equivalent to adding a regularizer to the attractive component of the loss function (i.e., $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$). Beyond attracting distant pairs, the modification promotes intermixing among points, which can assist convergence from random initialization in a manner analogous to early exaggeration in t -SNE. In [13], we use the simplest linear correction, although other functional forms, such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$), may also be suitable.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

The composite shape is designed to remove distortions introduced by f_a^M by reverting, after the initial phase, to the original UMAP attraction profile. We next examine how the modified and composite attraction shapes affect dimensionality reduction from random initialization.

We first generated a PCA-initialized embedding of the MNIST dataset (Fig. 2(a)). For each attraction shape, we then produced Gaussian-randomly initialized embeddings and repeated the experiment 100 times. To quantify consistency, we use Procrustes analysis (Gower [1975]), which aligns two point clouds under scaling, translation, rotation, and reflection (for details, see Appendix D.1). Specifically, we align each randomly initialized embedding to the PCA-initialized reference and characterize their separation using Procrustes distance (p_d). Figure 2(b) shows the four embeddings with the lowest p_d ; although their cluster shapes are consistent, their placements are not. In contrast, outputs from the modified and composite attraction shapes (Figs. 2(c) and (d), respectively) exhibit improved consistency in cluster placement.

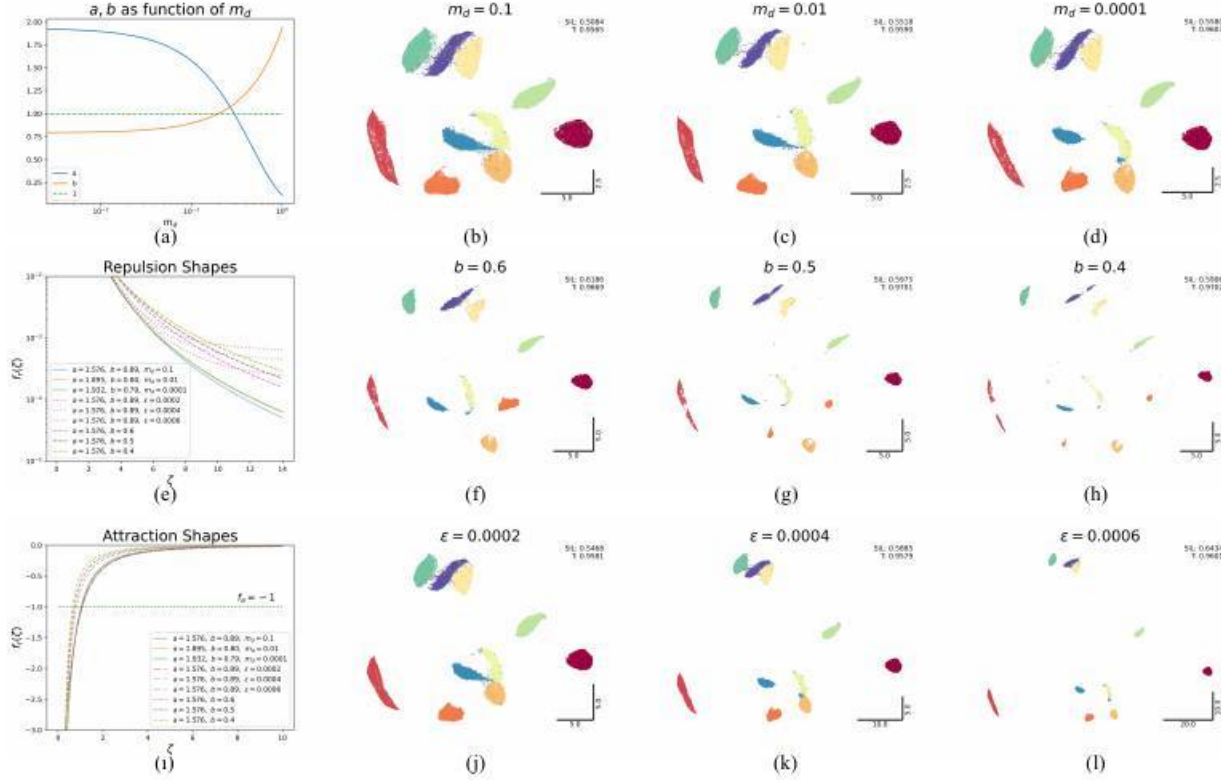


Figure 3: Control of inter-cluster distances on the MNIST dataset. (a) Estimating a, b by varying the low-dimensional distance m_d constrains the explored parameter region. (b-d) UMAP outputs obtained with m_d equal to 0.1, 0.01, and 0.0001, respectively, yield little improvement in cluster compactness. (e) Repulsion shapes under different parameters. (f-h) Explicitly increasing repulsion by varying b produces more compact clusters and reveals clusters absent under the default setting. (i) Attraction shapes induced by parameter variation. (j-l) Adding a small positive value (ϵ) to the repulsion shape increases inter-cluster distance.

To further quantify placement consistency, we construct the Procrustes matrix: diagonal entries are sorted by p_d from the PCA-initialized mapping, whereas off-diagonal entries give p_d between pairs of randomly initialized mappings (for details, see Appendix D.1). This representation is analogous to a similarity matrix (Foote, 1999). Embeddings produced by the default UMAP attraction shape are not mutually similar (Fig. 2(f)), whereas those produced by the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes exhibit strong mutual similarity. This pattern indicates that UMAP encodes a unique structure independent of initialization, even though recovering that structure in low dimension can fail when attraction at longer distances is too weak.

5.2 Cluster Formation and Compactness

The most prominent user-facing parameter controlling cluster formation in UMAP is the minimum-distance parameter m_d (through Eq. 2). However, varying m_d constrains the range of a and b values that can be explored (Fig. 3(a)), and reducing m_d therefore often yields embeddings with little additional compactness benefit (Figs. 3(b-d)). The central limitation is that m_d has only a restricted effect on the repulsion shape (Fig. 3(e)). A more direct alternative is to vary a and b themselves. Decreasing a increases repulsion, but because it decreases attraction even more rapidly, it also exacerbates near-sightedness. Decreasing b , by contrast, provides more effective control (Figs. 3(e,i)). Figures 3(f-h) show that setting b to 0.6, 0.5, and 0.4, respectively, increases inter-cluster distance and partitions clusters that were previously merged. This partitioning reflects the increasingly heavy-tailed kernel induced by smaller b values, since heavy-tailed kernels tend to produce smaller and more distinct clusters (Van der Maaten and Hinton, 2008; Kobak et al., 2019; see Kobak et al., 2019 for an account of how varying b modulates the UMAP tail. Although this manipulation separates all ten MNIST labels into distinct clusters, it does not by itself isolate attraction from repulsion, because changing b amplifies repulsion more than changing m_d while simultaneously reshaping the attraction profile.

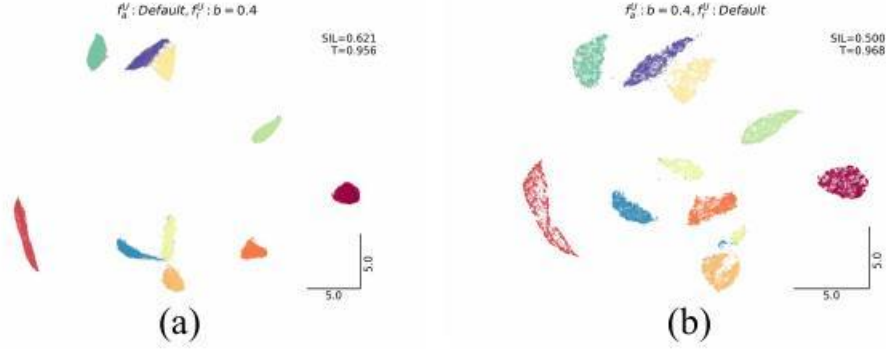


Figure 4: MNIST embeddings obtained by independently exchanging force shapes: (a) default attraction with the repulsion shape induced by $b = 0.4$, and (b) default repulsion with the attraction shape induced by $b = 0.4$. The former preserves the default UMAP cluster identities while increasing compactness, whereas the latter reveals new intra-cluster structure and forms additional clusters.

To quantify repulsion independently, we hold attraction fixed and vary only the repulsive component. Specifically, we modify the repulsion shape by adding a small positive value (ε):

$$f_r^M = f_r^U + \varepsilon, \quad (15)$$

while keeping a and b fixed, which effectively adds a regularizer to the repulsive term of the loss function (i.e., $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \varepsilon/2 \|y_i - y_j\|_2^2$). This modification produces stronger repulsion than is available through the preceding parameter variation (Fig. 3(e)). Figures 3(j-l) show that increasing ε enlarges inter-cluster distances. Nevertheless, the resulting clustering resembles the behavior obtained by varying m_d , and larger ε yields only a loose separation of all labels. Because ε leaves the attraction shape unchanged, while varying m_d effectively traces similar attraction shapes (Fig. 3(i)), these results suggest that cluster formation is governed predominantly by attraction.

To disentangle the source of cluster formation more directly, we alter either the attraction shape or the repulsion shape while keeping the other at its default, using the shape induced by $b = 0.4$ as the replacement (Fig. 4). With default attraction, increased repulsion produces clusters that are smaller than those of the original UMAP but does not reveal new structures or new cluster identities (Fig. 4(a)). Conversely, when attraction is increased through the $b = 0.4$ shape while repulsion remains default, the embedding exhibits additional structure within clusters, and several preexisting clusters split into smaller groups (Fig. 4(b)). These observations indicate that attraction drives cluster formation, whereas repulsion primarily increases compactness, either by shrinking clusters or by enlarging inter-cluster distances, depending on the repulsion shape. Appendix G.3 shows that LocalMAP embeddings exploit the same attraction–repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

Many dimensionality-reduction algorithms adopt the special parameterization $a = 1$ and $b = 1$, because this setting stabilizes, or bounds, the gradients and thereby simplifies optimization. Recently, Damrich et al. (2023) examined this regime for UMAP and proposed Neg- t -SNE, which combines gradient stability with more compact clustering even under a constant learning rate of 1. Parametric UMAP (Sainburg et al., 2021), by contrast, originally used the UMAP loss formulation, but later it adopted a numerically stable modified cross-entropy loss (Shi et al., 2023) based on a logsigmoid kernel. Analyzing these two developments shows that they reach the same formulation from different starting points, yielding the following observation:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

When $a = 1$ and $b = 1$, the UMAP attraction and repulsion shapes become $f_a^U = -2/(1+\zeta^2)$ and $f_r^U = 2/(\zeta^2(1+\zeta^2))$, respectively. The attraction shape is then bounded within $[-2, 0]$, with $f_a^U(0) = -2$ (Fig. 1(f)), whereas the repulsion shape remains essentially unchanged and remains unbounded as $\zeta \rightarrow 0$ (Fig. 1(g)). Because $f_a^U < -1$ in the small-distance limit, Proposition 4.1 together with the analysis in Section 5 implies that even this unity-parameter case still requires learning-rate annealing.

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Damrich et al. (2023) analyzed UMAP’s negative-sampling loss from the perspective of contrastive embedding (CE) and concluded that UMAP’s effective kernel is $1/\zeta^2$. Within the CE framework, NEG- t -SNE is obtained by replacing this kernel with $1/(1 + \zeta^2)$. Expressed in UMAP notation, this substitution yields the low-dimensional affinity $q_{ij}^N = 1/(2 + \zeta^2)$ and, consequently, the attraction and repulsion shapes $f_a^N = -2/(2 + \zeta^2)$, and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). The attraction shape is bounded within $[-1, 0]$ and satisfies Proposition 4.1 accordingly, any $\lambda \in [0, 1]$ contracts distances without inducing oscillatory expansion and contraction. NEG- t -SNE is therefore less sensitive to learning-rate annealing, and its clusters remain less fuzzy even when $\lambda = 1.0$ is held constant. Its repulsion shape is also bounded within $[0, 1]$ and does not diverge as $\zeta \rightarrow 0$, which reduces the tendency for randomly sampled points to leave clusters. Although Damrich et al. (2023) explained the disparity primarily through bounded NEG- t -SNE repulsion versus unbounded UMAP repulsion, our analysis in Fig. 1 shows that attraction is equally responsible for cluster stability and compactness. Additional illustrated discussion of NEG- t -SNE appears in Appendix F. Together with Fig. 4 this analysis suggests that attraction and repulsion shapes can be mixed across methods, for example by combining UMAP’s attraction shape with NEG- t -SNE’s repulsion shape; Appendix J provides experiments on UMAP, NEG- t -SNE, and PaCMAP.

7 Discussion and Conclusion

This work examined how attractive and repulsive forces jointly govern cluster formation in neighbor-embedding methods. Although attraction is conventionally described as pulling similar points together and repulsion as pushing dissimilar points apart, the specific mechanisms through which these forces generate cluster geometry have remained insufficiently characterized. By analyzing the shapes of the underlying force coefficients, we clarified a substantial portion of the dynamics responsible for cluster formation. The main text focuses on UMAP and NEG- t -SNE, but the resulting perspective is broadly applicable to dimensionality reduction, with additional algorithms discussed in Appendix G.

By treating attraction and repulsion coefficients as functional mappings, we characterized their respective shapes and derived conditions for contraction and expansion (Propositions 4.1 and 4.2). This characterization revealed a counterintuitive property of UMAP: at short distances, its nominally attractive force can induce repulsion, or distance expansion (Fig. 1(b)), so that neighboring points are pushed apart rather than pulled together. UMAP’s learning-rate schedule alleviates this phenomenon by progressively reducing the problematic contraction threshold.

We also analyzed initialization effects in the low-dimensional embedding. Similar points that begin far apart experience weak attraction and may therefore fail to contract adequately. The force-shape formalism provides a direct way to strengthen attraction at larger distances by adding terms to the attraction shape, and this modification yields outputs that are substantially more consistent under random initialization. These results support the view that UMAP encodes a reproducible underlying structure, while numerical strategies primarily help the optimization reach that structure more reliably.

Analysis of the repulsion shape showed that stronger repulsion enlarges inter-cluster distances. By adding a small positive value directly to the repulsion shape, we obtained a level of repulsion not accessible through the conventional adjustment of the minimum-distance parameter m_d .

Finally, we clarified the relationship between NEG- t -SNE and UMAP. In the special case $a = 1$ and $b = 1$, the principal distinction is that default NEG- t -SNE supplies bounded attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) shapes, so the contraction and expansion propositions are satisfied across the relevant learning-rate range. Consequently, the method is less sensitive to fixed learning rates in $[0, 1]$. Allowing a and b to vary in NEG- t -SNE, however, reintroduces numerical challenges analogous to those in UMAP. We also established the equivalence between NEG- t -SNE and Parametric UMAP.

Together, these insights into attraction–repulsion dynamics provide new tools for optimizing dimensionality-reduction algorithms and for interpreting the visual structures they produce. Because dimensionality reduction is closely connected to contrastive learning (Damrich and Hamprecht, 2021; Hu et al., 2023), the same force-shape perspective may also inform representation-learning methods. Overall, the analysis aims to make embeddings, and the scientific interpretations drawn from them, more principled, consistent, and reliable, while motivating future work on force-aware algorithm design.

Software and Data

All datasets used in this paper are publicly available. The MNIST dataset is available at yann.lecun.com. Fashion-MNIST is available at github.com.

Single-cell transcriptomes data is available at github.com

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Appendix

Appendix [A](#) provides the derivations required for the UMAP attraction and repulsion shapes. Building on the main-text discussion, Appendix [B](#) revisits constant learning rates, and Appendix [C](#) examines the implications of improved attraction and repulsion for large language model embeddings. Appendix [D.1](#) formally defines the Procrustes distance and Procrustes matrix, while Appendix [E](#) extends the random-initialization experiments to additional datasets. Appendix [F](#) expands the comparison between NEG- t -SNE and UMAP, and Appendix [G](#) analyzes alternative dimensionality-reduction methods. For completeness, Appendix [H](#) describes construction of the high-dimensional graph, Appendix [I](#) reports detailed results for varying λ_a and λ_r (from Fig. [1](#)) across UMAP, NEG- t -SNE, and PaCMAP on multiple datasets, Appendix [J](#) studies the effect of mixing force shapes from different algorithms, and Appendix [K](#) supplies implementation details.

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. Subtracting Eq. [\(7\)](#) from Eq. [\(8\)](#) and taking the norm gives

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

The distance contracts precisely when $|1 + 2\lambda f_a| < 1$, which is equivalent to the condition

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. Subtracting Eq. [\(9\)](#) from Eq. [\(10\)](#) and taking the norm yields

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance expands whenever $|1 + \lambda f_r| > 1$, equivalently when

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

Because of the definition of f_r , the latter condition is sufficient. □

A.3 Attraction Shape:

To derive the attraction shape, we use the general low-dimensional affinity $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, which reduces to UMAP when $\gamma = 1$ and to NEG- t -SNE when $\gamma = 2$. The corresponding attractive force is

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

After defining $\zeta = \|y_i - y_j\|_2$, the scalar coefficient gives the attraction shape

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

A.4 Condition for strictly increasing f_a^U :

The distance dependence of f_a^U is characterized by differentiating the shape:

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

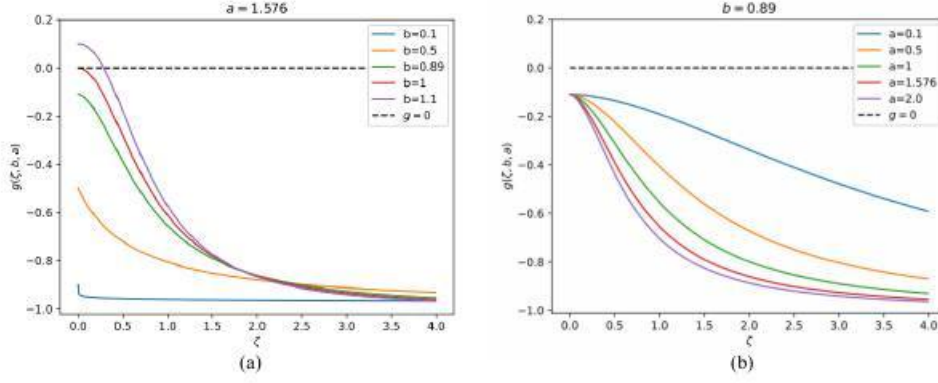


Figure 5: Values of $g(\zeta, b, a)$ when (a) a is fixed at 1.576 and (b) b is fixed at 0.89.

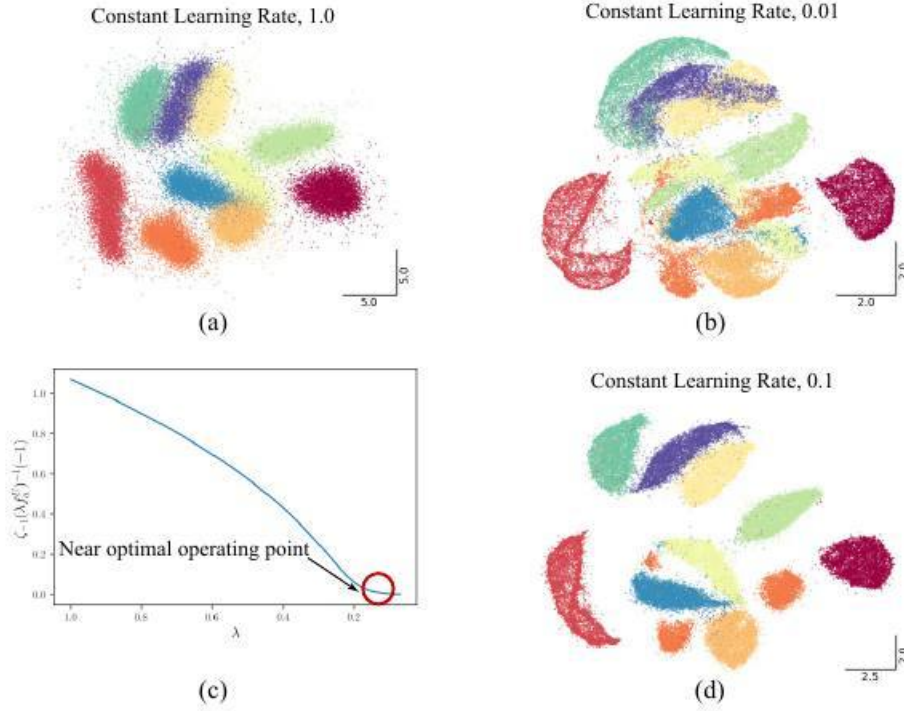


Figure 6: Effect of a constant learning rate on embeddings. (a) With an excessively high learning rate ($\lambda = 1.0$), embeddings become diffuse because ζ_{-1} is large. (b) With an excessively low learning rate ($\lambda = 0.01$), clusters fail to form because attraction and repulsion are both too weak. (c) ζ_{-1} decreases nonlinearly as the learning rate decreases, so optimization must reduce ζ_{-1} while retaining sufficient attraction and repulsion. (d) Distinct, compact clusters emerge at the near-optimal constant learning rate $\lambda = 0.1$.

This derivative yields the strict-monotonicity condition ($\frac{df_o^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. Using the derivative and asymptotic behavior of g , the inequality holds for $0 < b \leq 1$. Figure 5 displays g for different values of b and a , and shows that once b exceeds 1, inequality (23) no longer applies.

A.5 Repulsion Shape:

Using the same general low-dimensional affinity, the repulsive force is

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The scalar coefficient therefore gives the repulsion shape

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

In general, this shape satisfies $f_r > 0$.

A.6 Loss functions due to modified attraction and repulsion shape:

With the modification in Eq. (13), the attractive-term cost function becomes

$$- \int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

whereas the repulsive term induced by Eq. (15) is

$$- \int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases, the additional norm-based term functions as a regularizer. Expressing the modification directly at the level of attraction and repulsion shapes, however, makes its interpretation immediate: for attraction, the term increases far-sightedness, whereas for repulsion, it adds a constant repulsive coefficient.

B More on UMAP’s Learning Rate

As discussed in Sections 5 and 6 UMAP is generally understood to require learning-rate annealing (Fig 6). Section 5 explains this requirement through the minimum distance for contraction (ζ_{-1}), showing that reducing this quantity through annealing yields compact clusters. Section 6 then compares the attraction shapes of UMAP (with $a = 1.0$ and $b = 1.0$) and Neg- t -SNE, demonstrating that Neg- t -SNE tolerates a constant learning rate of 1.0 more effectively because its attraction shape lies within $[-1, 0]$, whereas UMAP’s lies within $[-2, 0]$. By the same reasoning, UMAP can tolerate a constant learning rate of 0.5 when its effective attraction shape is confined to $[-1, 0]$ (Figs. 11(f,h)).

Nevertheless, across a wide range of a and b values, embeddings remain better when the learning rate is annealed. The reason is that the algorithm ultimately benefits from reducing ζ_{-1} to zero, which occurs only as the learning rate approaches zero. If the learning rate remains too high, the embedding becomes diffuse (Fig 6(a)); if it is too low from the outset, attraction and repulsion are both too weak to produce clear clusters (Fig 6(b)). The ζ_{-1} versus λ curve (Fig 6(c)) indicates that $\lambda = 0.1$ is near optimal, because it keeps ζ_{-1} low while preserving substantial attraction and repulsion. With this constant learning rate, the embedding forms compact clusters with clear boundaries (Fig 6(d)).

C Large Language Model Embedding

Stronger attraction and repulsion can also improve embeddings in large language model feature spaces. To demonstrate this, we randomly selected 300,000 PubMedBERT (Gu et al., 2021) feature samples with known labels from the PubMed dataset (González-Márquez et al., 2024). Default UMAP produces a crowded structure in which labels, shown as colors, overlap substantially (Fig. 7(a)); without labels, the absence of explicit clusters would make the embedding difficult to interpret. By setting b to 0.6, as motivated in Section 5.2 we increase repulsion while simultaneously facilitating cluster formation through improved attraction. The resulting mapping is explorable even without explicit labeling (Fig. 7(b)). This improvement is reflected quantitatively in k-NN accuracy ($k = 10$): on a hold-out set of 10,000 points, reducing b to 0.6 increases k-NN accuracy from 49.2% to 55.48%.

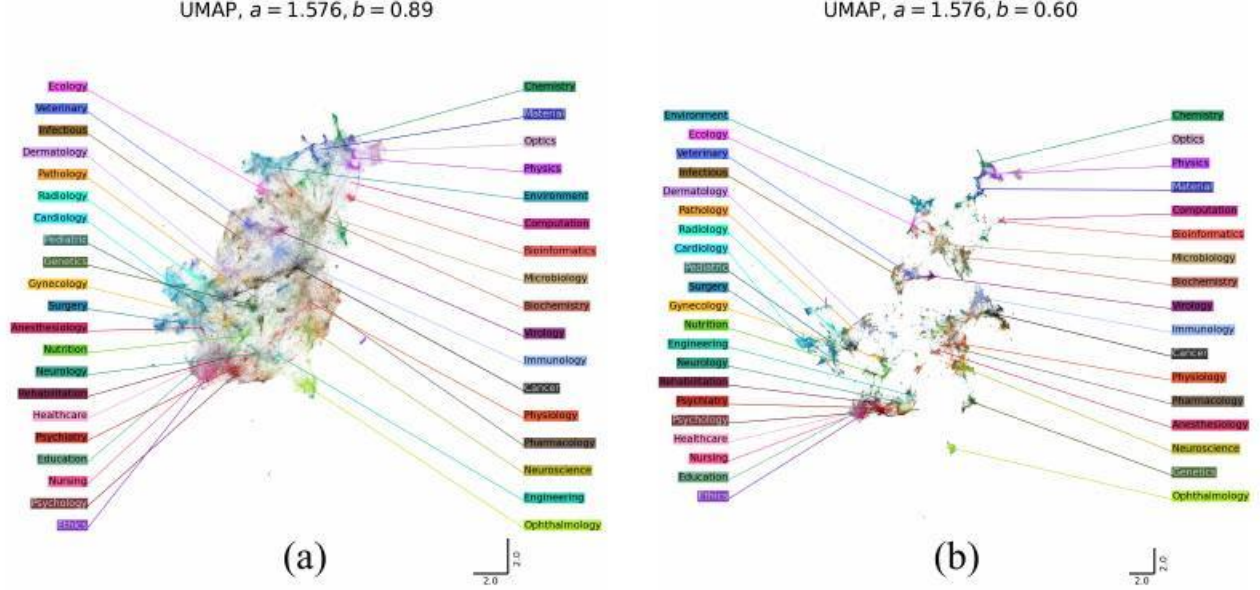


Figure 7: Visualization of 300K abstracts from the PubMed dataset. (a) Default UMAP ($a = 1.576, b = 0.89$) produces a crowded mapping in which labels overlap, whereas (b) UMAP with strengthened repulsion and attraction ($a = 1.576, b = 0.60$) yields clearer cluster formation and non-overlapping labels. Both mappings are labeled to facilitate interpretation.

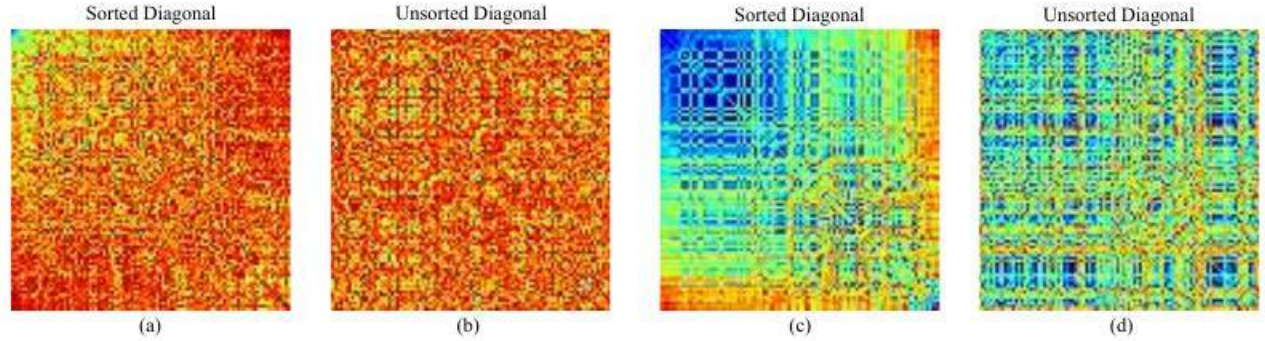


Figure 8: Effect of diagonal sorting on Procrustes-matrix visualization. (a) Procrustes matrix reproduced from [2]f for the default UMAP attraction shape, with diagonal entries sorted by p_d from a PCA-initialized sample embedding. (b) The same data as in (a), but with the diagonal unsorted, or randomly sorted. (c) Procrustes matrix reproduced from [2]g for the modified attraction shape, again with diagonal entries sorted by p_d from a PCA-initialized sample embedding. (d) The same data as in (c), but with the diagonal unsorted.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower [1975]) measures the similarity between two point clouds $\{x\}$ and $\{y\}$ after allowing linear transformations such as translation, scaling, and rotation. Operationally, we hold the first point cloud fixed and transform the second until the two are maximally aligned. If $\{y'\}$ denotes the transformed version of $\{y\}$ that achieves this alignment, the Procrustes distance is

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

The Procrustes distance is a linear measure that has proved useful across several embedding-analysis settings (McInnes et al. [2018]; Islam and Fleischer [2022]; Kotlov et al. [2024]).

Here, we use Procrustes distance to quantify embedding consistency under random initialization. Let $\{x\}_p$ denote a reference embedding, obtained through PCA initialization in our experiments, and let $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ denote N embeddings obtained from random initialization. Embedding similarity is summarized by the mean and standard deviation of the strictly lower-triangular entries of the matrix P (reported as mean $\pm$ std in Figs. 2, 9, and 10), with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indices of X_r can be ordered so that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, in which case the diagonal entries of the Procrustes matrix are

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal entries are

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Although diagonal sorting contributes little numerically, it substantially improves visual interpretation by revealing latent self-similarity among embeddings. In Fig. 8(a), for example, sorting the diagonal causes mutually similar embeddings to cluster in the upper-left portion of the matrix. Under the modified attraction shape, the set of mutually similar embeddings expands, as indicated by the larger blue region in Fig. 8(c), suggesting the presence of a metastable point in the embedding algorithm. When the diagonal is unsorted, this region disappears and the visual evidence of embedding similarity is obscured (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski 2001) quantifies how well local neighborhoods are preserved after dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ denotes the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. In practice, k is often set to 5, thereby assessing preservation of each point's five nearest neighbors. For computational efficiency, all reported trustworthiness values use 10,000 randomly sampled indices, and comparisons across embeddings use the same sampled indices.

D.3 Silhouette Score

Although the silhouette score (Rousseeuw, 1987) was designed to assess clustering algorithms, we use it here to evaluate label separation in embeddings, namely how well ground-truth labels are separated by the learned geometry. The premise is that neighbor-embedding algorithms naturally produce clusters and should separate labels as clearly as possible. For a point y_i in a point cloud $\{y\}$, let a_i denote the mean distance from y_i to other points with the same label, and let b_i denote the smallest mean distance from y_i to points in any other label. The pointwise silhouette is then

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This quantity lies in $[-1, 1]$: values near 1 indicate that y_i is well embedded within its own label cluster, values near 0 suggest a boundary point, and values near -1 indicate failed label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We compute the silhouette score on the full embedding, without random sampling, using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

The main text (Section 5.1) reports random-initialization results for MNIST. Here, we repeat the same experiment on Fashion-MNIST (FMNIST) (Xiao et al., 2017) and single-cell transcriptomes (Macosko et al., 2015) data (Fig. 9 and 10, respectively). The central conclusion is unchanged: modified and composite attraction shapes that increase long-distance attraction substantially improve reconstruction consistency.

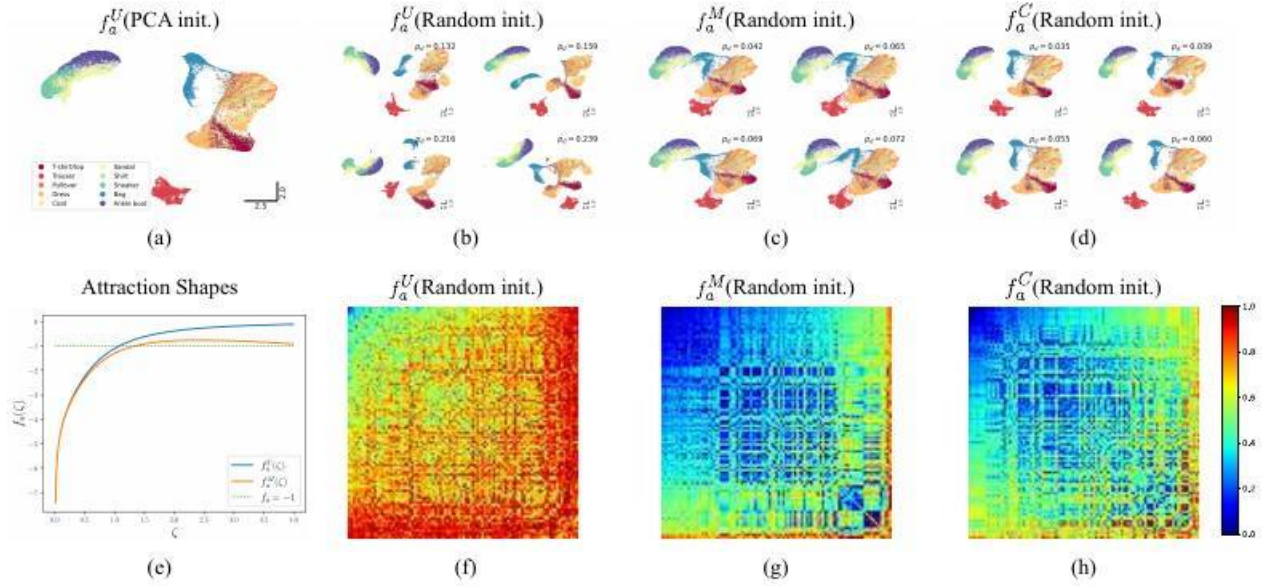


Figure 9: Effect of random UMAP initialization under different attraction shapes on FMNIST data. (a) PCA-initialized mapping used as the reference. (b-d) Four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices obtained from 100 runs using (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes.

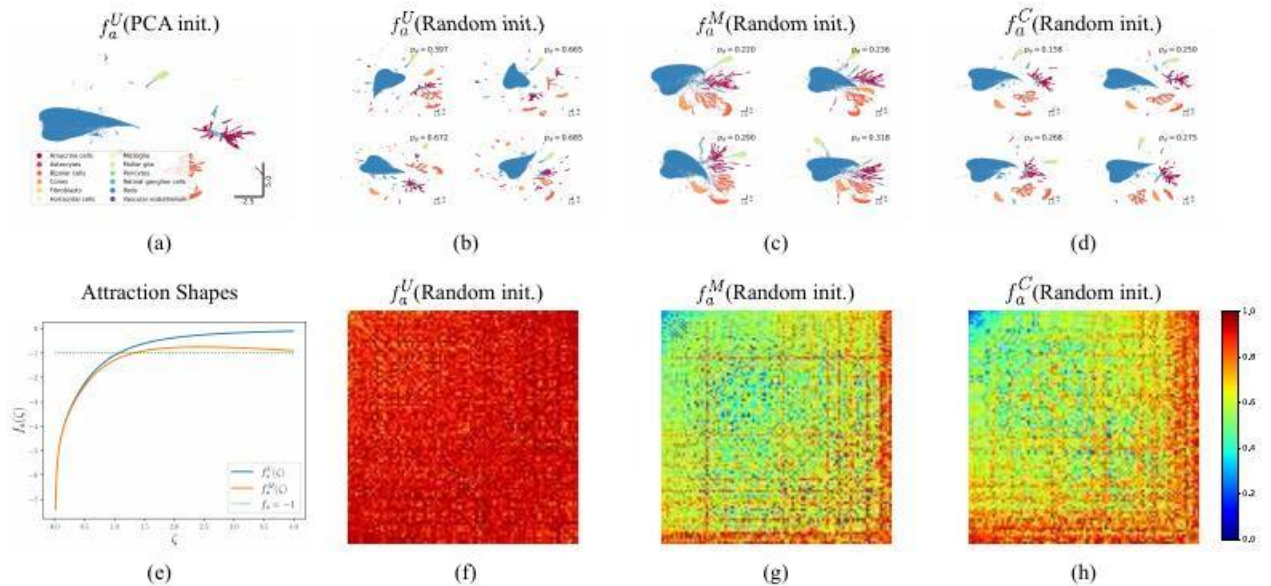


Figure 10: Effect of random UMAP initialization under different attraction shapes on single-cell transcriptomes data. (a) PCA-initialized mapping used as the reference. (b-d) Four mappings with the lowest Procrustes distance (p_d) from the embedding in (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices obtained from 100 runs using (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes.

F More on NEG- t -SNE

F.1 Comparison to UMAP

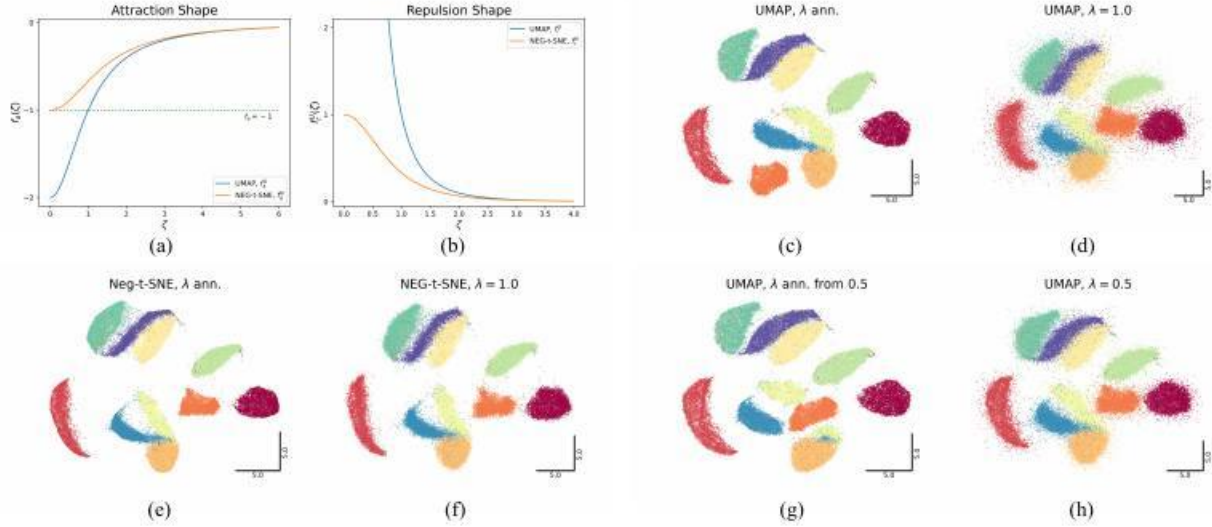


Figure 11: Learning-rate sensitivity of UMAP and NEG- t -SNE on the MNIST dataset. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG- t -SNE. (c,d) UMAP is highly sensitive to the learning rate λ because $f_a^U < -1$ as the separation distance ζ decreases, making clusters fuzzy without annealing. (e,f) NEG- t -SNE is less sensitive to λ because $f_a^N \in [-1, 0]$ for all distances, so clusters remain less fuzzy even without annealing. (g,h) Restricting f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ similarly reduces sensitivity to λ .

Figure 11 compares the shapes of UMAP and NEG- t -SNE alongside several MNIST embeddings. When the learning rate is annealed, UMAP and NEG- t -SNE produce similar outputs (Figs. 11(c,e)); under a constant learning rate of 1, however, UMAP yields a fuzzy structure, whereas NEG- t -SNE produces substantially cleaner boundaries (Fig. 11(d,f)). The discussion in Section 6 implies that constraining f_a^U to $[-1, 0]$ should reduce fuzziness under fixed λ , a pattern also observed in Fig. 11(h,i), where UMAP achieves better embedding and clustering when the attraction learning rate (λ_a) is $\lesssim 0.5$. A direct way to impose this constraint is to initialize λ at 0.5, which satisfies Proposition 4.1 for all ζ . The embeddings in Figs. 11(g) and (h) confirm that the resulting clusters resemble those of NEG- t -SNE and, at constant $\lambda = 0.5$, exhibit sharper boundaries and less fuzziness than before (Fig. 11(d,h)). A few points remain outside the clusters because of UMAP’s repulsion shape, a behavior addressed by NEG- t -SNE.

We can also introduce the parameters a and b into NEG- t -SNE, which essentially yields the Parametric UMAP formulation (see Proposition 6.1). The affinity becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2 + a\zeta^{2b}}$, and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1 + a\zeta^{2b})(2 + a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes diverge as $\zeta \rightarrow 0$, meaning that variable a and b introduce numerical challenges in NEG- t -SNE analogous to those in UMAP. A notable distinction is that, relative to UMAP, the NEG- t -SNE attraction shape attains a lower minimum distance (ζ_{-1}). Although this may improve cluster formation, the shape also decays to zero more rapidly as distance increases, thereby increasing near-sightedness and potentially weakening long-range attraction.

F.2 Comparison to Parametric UMAP

Parametric UMAP was initially trained with the original UMAP objective (Sainburg et al. 2021), but subsequent work adopted a numerically stable modified cross-entropy loss based on the log-sigmoid function (Shi et al. 2023). This modification makes Parametric UMAP equivalent to Neg- t -SNE (Damrich et al. 2023) (Proposition 6.1); the equivalence is shown below.

Proof of Proposition 6.1

Proof. Under this modification, the kernel function becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

The repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Together, these terms recover NEG-t-SNE with explicit parameters a and b , whereas standard Neg- t -SNE fixes both parameters at 1. $\square$

G Alternate Dimensionality Reduction Algorithms

The alternative algorithms considered here use the same low-dimensional kernel as UMAP when $a = 1$ and $b = 1$:

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

This section first examines TriMap (Amid and Warmuth, 2019). Although TriMap relies on triplets rather than pairwise interactions, it provides a useful bridge toward analyzing attraction and repulsion dynamics that are more coupled than those in UMAP. We then discuss Pairwise Controlled Manifold Approximation (PaCMap) (Wang et al., 2021) and its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al., 2025), both of which modify TriMap-style objectives into pairwise forms. The section subsequently treats t -SNE (Van der Maaten and Hinton, 2008), briefly considers SNE (Hinton and Roweis, 2002), which uses an alternative kernel, and concludes with multidimensional scaling (Borg and Groenen, 2007).

G.1 TriMap

TriMap (Amid and Warmuth, 2019) optimizes the low-dimensional embedding through a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the triplet weight, y_j belongs to the high-dimensional k -nearest-neighbor set of y_i , and y_k is a far-away point. Minimizing this objective is expected to attract y_i and y_j while repelling y_i and y_k . The resulting update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the nearest-neighbor distance, $\zeta_2 = \|y_i - y_k\|_2$ is the distance to the faraway point, and f_a^T and f_r^T are the TriMap attraction and repulsion shapes, respectively. Unlike in UMAP, the attractive and repulsive components are not separable, and the shapes depend on two distance variables, making them two-dimensional. The attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

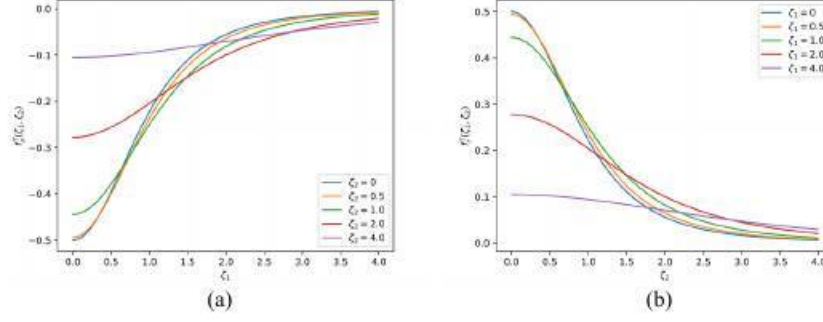


Figure 12: TriMap force shapes: (a) attraction shapes for different ζ_2 values and (b) repulsion shapes for different ζ_1 values.

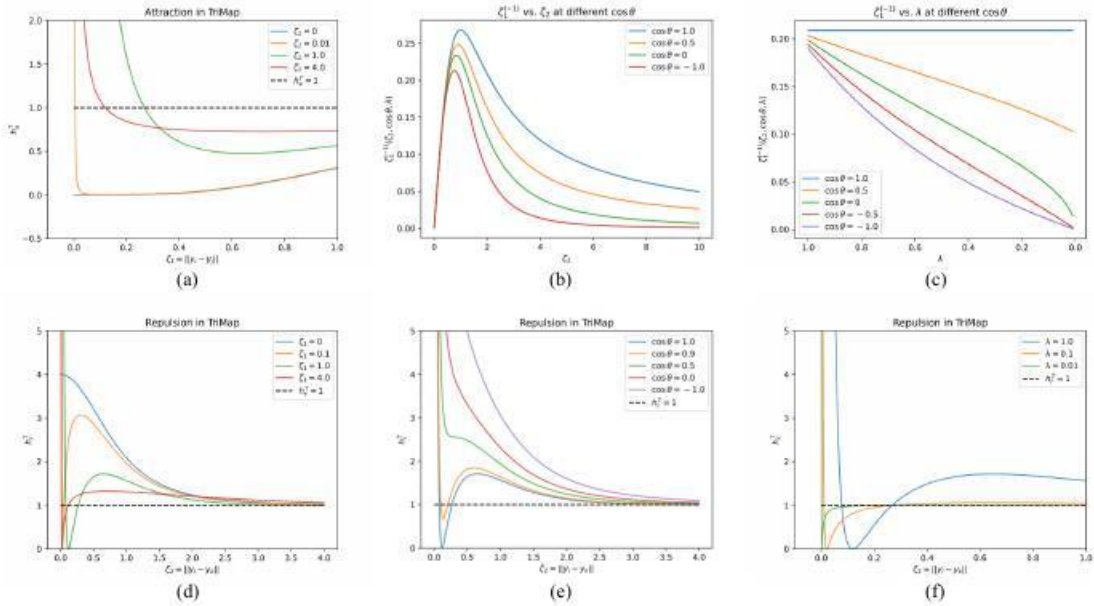


Figure 13: Attraction and repulsion behavior in TriMap. (a) h_a^T as a function of ζ_1 for different ζ_2 values, with values below the dotted line indicating attraction. As in UMAP, nearest-neighbor pairs (y_i, y_j) can experience both attraction and repulsion. (b,c) Unlike UMAP, the minimum distance for contraction ($\zeta_1^{(-1)}$) varies with (b) ζ_2 and (c) λ , while $\cos \theta$ regulates the range of possible values. (d-f) TriMap repulsion as (d) ζ_1 , (e) $\cos \theta$, and (f) λ vary. Values above (below) the dotted line indicate repulsion (attraction). Unlike UMAP repulsion, TriMap repulsion can induce both attraction and repulsion. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

The TriMap attraction and repulsion shapes (Fig. 12) exhibit trends similar to those of UMAP. However, because the minimum value of the attraction shape is -0.5 , TriMap does not produce position flipping through attraction alone. Since Eqs. (45)-(47) do not decouple attractive and repulsive terms, Propositions 4.1 and 4.2 do not apply directly. Focusing first on attraction, we obtain the following result.

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46), $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Substituting this expression into Eq. (51) yields the desired inequality. $\square$

Inequality (50) depends jointly on ζ_1 , ζ_2 , $\cos \theta$, and λ . Thus, the nearest-neighbor distance $\zeta_1 = \|y_i - y_j\|_2$, whose contraction is desired, is coupled to additional geometric variables. Figure 13(a) shows the resulting attraction behavior for different ζ_2 values with $\cos \theta = 1$ and $\lambda = 1$. Values below the dotted line indicate attraction, and hence contraction of ζ_1 , whereas values above the line indicate repulsion, and hence expansion of ζ_1 . The intersection between the dotted line and h_a^T defines the minimum distance for contraction ($\zeta_1^{(-1)}$), analogous to UMAP's ζ_{-1} , which we define as

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

For most parameter settings, $\zeta_1^{(-1)}$ is finite and positive (Fig. 13(b,c)). Consequently, TriMap can exhibit UMAP-like behavior and may require learning-rate annealing or an analogous mechanism; in the TriMap implementation, the authors use the delta-bar-delta method (Jacobs [1988]) under appropriate initialization.

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47), $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Substitution into Eq. (55) gives inequality (54). $\square$

Inequality (54) also depends on ζ_1 , ζ_2 , $\cos \theta$, and λ , but here the quantity of interest is expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) show how the repulsion behavior changes as the other variables vary. Values above the dotted line indicate repulsion, or distance expansion, whereas values below the line indicate attraction. The striking difference from UMAP is that TriMap repulsion can induce contraction rather than expansion. Because this anomaly occurs at small distances, appropriate initialization and triplet selection can avoid it.

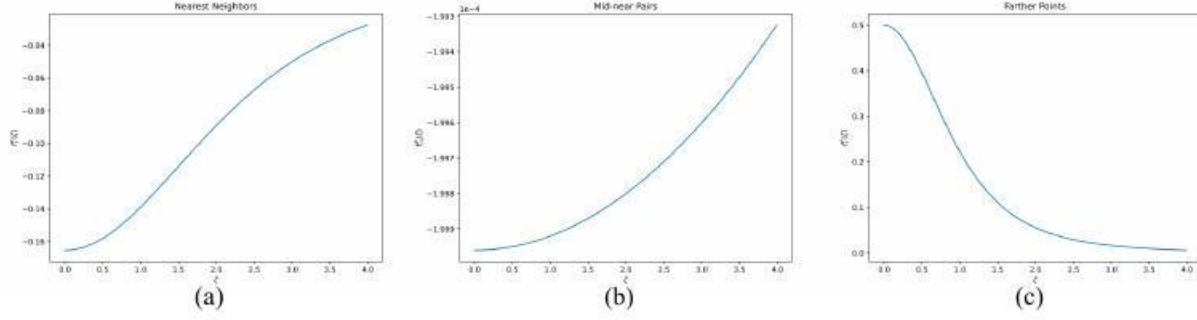


Figure 14: PaCMAP attraction and repulsion shapes. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near points, with values on the order of 10^{-4} . (c) Repulsion shape for farther pairs. $\lambda = 1$ in all panels.

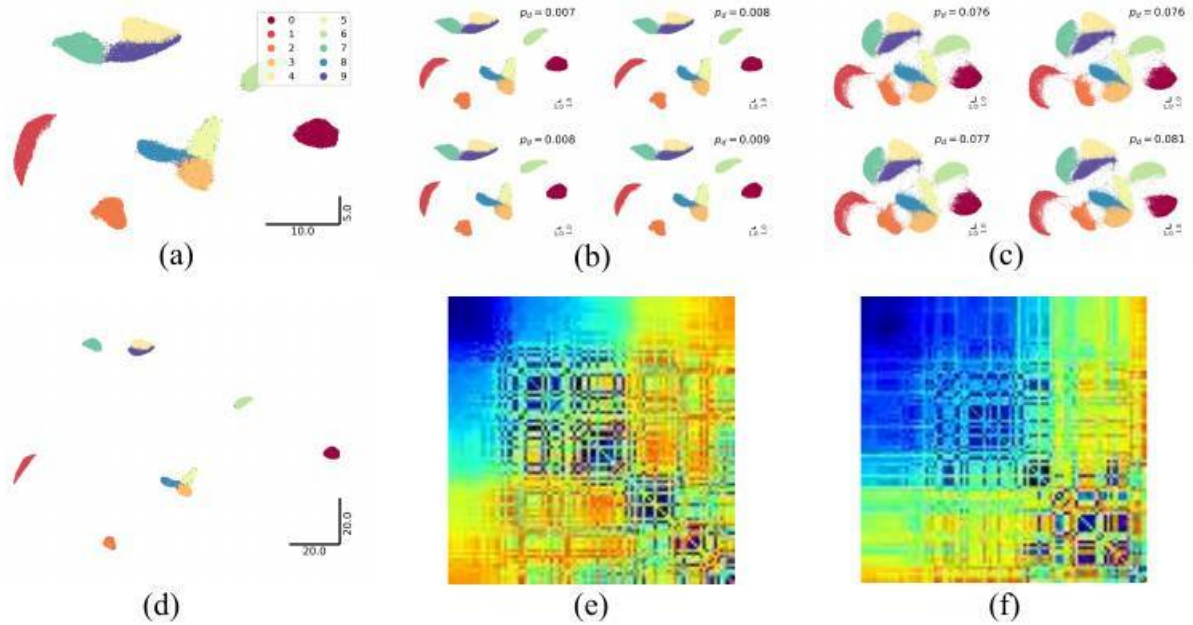


Figure 15: PaCMAP behavior under different conditions. (a) PaCMAP embedding of MNIST with PCA initialization. (b,c) Four randomly initialized samples that best match (a) for (b) original PaCMAP and (c) PaCMAP with modified attraction shape, $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) PaCMAP embedding of MNIST with modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) original PaCMAP (0.51 ± 0.22) and (f) PaCMAP with modified attraction shape (0.43 ± 0.22). As in UMAP, stronger long-distance attraction improves consistency, whereas stronger repulsion yields smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al. 2021) optimizes the low-dimensional embedding across multiple neighborhood scales. Its loss function is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} weight nearest-neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (details in Appendix H). The first two terms induce attraction, whereas the final term induces repulsion. The loss can be viewed as a modified version of TriMap’s triplet loss: in the attractive terms, PaCMAP replaces TriMap’s affinity for distant points with constants ($1/10$ for nearest neighbors, $1/10000$ for mid-near pairs, and 1 for farther points). Consequently, the loss separates into three terms and decouples the update equations. The nearest-neighbor

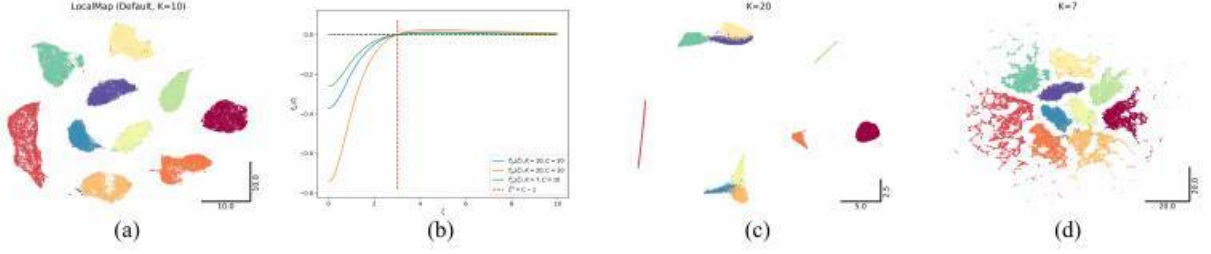


Figure 16: LocalMAP behavior on MNIST data. (a) Default embedding. (b) Attraction–repulsion shape for nearest neighbors, where values below (above) the dotted line indicate attraction (repulsion). The transition from attraction to repulsion occurs when ζ crosses $\sqrt{C-1}$; following the LocalMAP implementation, we use $C = 10$. (c) With large K ($= 20$), repulsion dominates and clusters become compact. (d) With small K ($= 7$), attraction dominates and existing clusters break apart.

updates are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

of the mid-near pairs are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and of the farthest pairs are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, and $\zeta_3 = \|y_i - y_l\|_2$ denote distances; f_a^P and f_m^P are the attraction shapes for nearest-neighbor and mid-near pairs, respectively; and f_r^P is the repulsion shape for farthest pairs. The corresponding functional forms are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

f_a^P and f_m^P satisfy Proposition 4.1 while f_r satisfies Proposition 4.2 (Fig. 14). PaCMAP attraction is relatively weak compared with UMAP, but it remains effective across a broad range of learning rates, in part because optimization is modulated by Adam (Kingma and Ba, 2015); with $w_{NB} = 3$, the maximum attraction remains below 0.5, preventing flips during attractive updates. PaCMAP typically initializes the embedding inside a small low-dimensional sphere (for example, on the order of 10^{-3} in (Wang et al., 2021)) and then relies on repulsion to separate clusters. Overall, it largely recovers UMAP’s clustering properties, especially on MNIST, while improving ordering. Its consistency under random initialization is better than UMAP (Fig. 15 b,e) and can be improved further through modified attraction (Fig. 15 c,f). Adding a small value to the repulsion shape increases embedding compactness by enlarging inter-cluster distances.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al., 2025) is an extension of PaCMAP. A defining feature of LocalMAP is its separation of all 10 MNIST clusters (Fig. 16 (a)), a behavior resembling the patterns in Figs. 3 (g,h,j,k) and Figs. 4 (a,b). We therefore use LocalMAP to examine how attractive and repulsive forces jointly affect cluster compactness and connectedness.

LocalMAP first applies PaCMAP and then performs additional attraction-focused optimization to decouple selected clusters. It minimizes the loss

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term combines attractive and repulsive behavior for the same pair: in one regime the term induces attraction, and in the other it induces repulsion. The second term is identical to the corresponding PaCMAP term, except that further pairs are resampled every few iterations. We therefore analyze only the first, nearest-neighbor term. The updates are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_j^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

The update contracts distances when $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$ (by Proposition 4.1). When $\zeta^2 > C-1$, $f_{ar} > 0$, and Proposition 4.2 implies expansion. The parameters K and C determine whether attraction or repulsion dominates. In the LocalMAP implementation, both C and K are set to 10 (Fig. 16(b)), and the attraction strength exceeds that of PaCMAP (Fig. 14(a)). Consequently, when $\zeta > 3$, nearest-neighbor pairs experience repulsion, causing bridge pairs between two clusters to separate. The parameter K acts as a force-scaling coefficient that regulates this separation.

Using Proposition 4.1, the condition $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ gives the largest permissible K , and hence the strongest possible attraction under f_{ar}^L , without flipping pair placements. Avoiding such flips is desirable during LocalMAP optimization because it preserves the ordering inherited from PaCMAP; otherwise, cluster separation may be inhibited. This condition reduces (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, which for $C = 10$ and $\lambda = 1$ gives $K \lesssim 13.44$. For larger values, $K \gtrsim 13.44$, repulsion dominates and clusters become more compact, but the LocalMAP objective fails because the bridge between clusters persists (Fig. 16(c) for $K = 20$). Conversely, as K decreases, attraction dominates because repulsion is too weak, and the embedding breaks existing clusters apart (Fig. 16(d) for $K = 7$), mimicking the behavior in Fig. 4(b).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton [2008]) optimizes pairwise relationships through the loss

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the pair weight. The original t -SNE implementation considers all pairs, rather than only nearest neighbors. The loss decomposes into attractive and repulsive components as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term induces attraction and the second induces repulsion. The attractive term is identical to the UMAP attractive term when $a = 1$ and $b = 1$, making it straightforward to compute; the repulsive term, however, is coupled across all pairs and is therefore computationally expensive. Applying the same principles used for UMAP yields the t -SNE update equations. Because the original t -SNE does not rely on a nearest-neighbor graph, the pair weight $w_{i,j}$, computed for all pairs, remains central in these updates. The attractive updates are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the t -SNE attraction shape and $\zeta_{i,j} = \|y_i - y_j\|_2$. The repulsive updates are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the t -SNE repulsion shape and $Z = \sum_{k \neq l} q_{k,l}$. These shapes take the forms

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1+\zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1+\zeta^2)^2}. \quad (77)$$

From the attractive updates in Eqs. (72)-(73), f_a^{t-SNE} satisfies Proposition 4.1 under the corresponding weighted condition (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and yields a minimum distance for contraction, ζ_{-1} . The repulsive update is coupled across all pairs and therefore lacks a simple one-dimensional relation to the repulsion shape. Instead, following the strategy used for TriMap, we obtain the following result.

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k, k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l, l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Substituting this expression into Eq. (79) gives the desired inequality. $\square$

Because the t -SNE condition for repulsion (Eq. 78) resembles the TriMap condition, the qualitative repulsion behavior is analogous. Thus, t -SNE repulsive forces can induce either attraction or repulsion depending on the configuration.

Because t -SNE forces are scaled by $w_{i,j}$ for attraction and by $\frac{w_{i,j}}{Z}$ for repulsion, both force magnitudes are typically smaller than their UMAP counterparts. The algorithm therefore commonly uses large learning rates, for example approximately 10^3 in the Open- t -SNE package (Poličar et al., 2024). Most t -SNE implementations also include an ‘early exaggeration’ phase, during which attractive forces are multiplied by a constant for the first several iterations. This phase draws together points that should be close but are initially far apart, thereby inducing far-sightedness; it can also separate points that are already very close ($\zeta_{i,j} \approx \zeta_{-1}$) but should not remain together. Consequently, early exaggeration is central to obtaining consistent t -SNE embeddings, especially under random initialization. When applied throughout optimization, the same mechanism makes t -SNE embeddings resemble UMAP embeddings more closely (Böhm et al., 2022).

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis, 2002) is among the earliest algorithms in this family. It follows the same loss structure as t -SNE (Eq. 71) but uses $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$, which yields the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which follows Proposition G.3 after replacing the t -SNE symbols with their SNE counterparts. The attraction shape is ill posed and therefore depends strongly on the learning rate (λ) and weights ($w_{i,j}$) to achieve contraction, leading to overlapping clusters even for small samples; this is the crowding problem (Van der Maaten and Hinton, 2008). t -SNE and subsequent methods mitigate this issue by replacing the Gaussian kernel with a heavy-tailed kernel, as in Eq. (43).

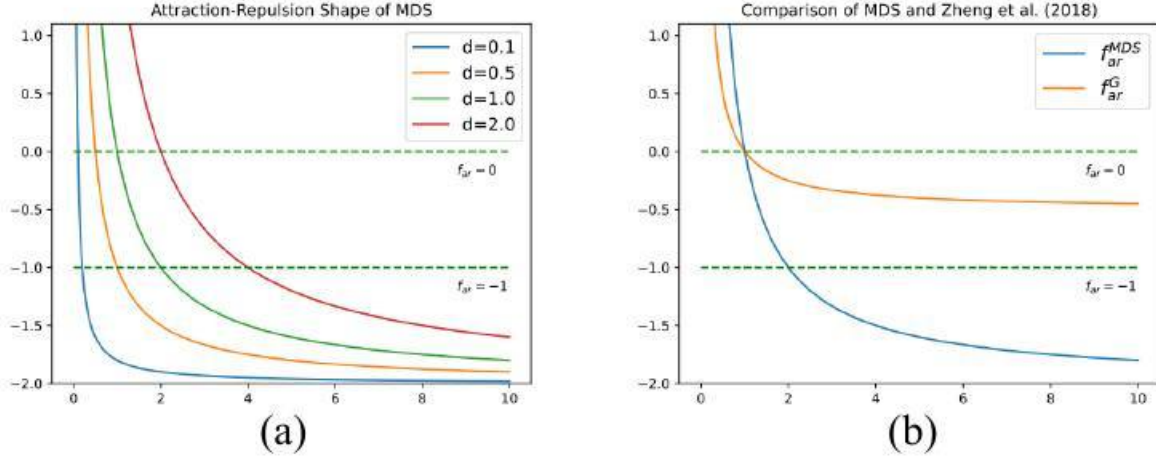


Figure 17: MDS attraction-repulsion behavior. (a) Attraction-repulsion characteristic of MDS using f_{ar} . (b) Comparison between the MDS attraction-repulsion characteristic and the method in (Zheng et al., 2018) for $d = 1.0$.

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) typically avoids gradient methods, which often fail to converge to high-quality mappings, and instead relies on stress majorization. Nevertheless, several works consider gradient-based variants (Kruskal, 1964; Zheng et al., 2018), and we briefly analyze that perspective here. In particular, Zheng et al. (2018) formulates a successful gradient-descent MDS algorithm for graph drawing. We begin with the cost function, ignoring the weight $w_{i,j}$ without loss of generality for their approach and absorbing it, for discussion, into the learning rate:

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss contains a single term for each pair and does not explicitly separate attraction from repulsion; rather, each term can induce both behaviors. Because the loss is separable across individual (i, j) components, a single-pair analysis is sufficient. The derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

Thus, the MDS attraction-repulsion shape is

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape is attractive when $\zeta > d$, and in this range it is bounded within $[-2, 0]$ (Fig. 17(a)). This interval is not ideal for convergence, however, because only values in $[-1, 0]$ contract; the remaining values expand, and values below -2 cause flipping followed by expansion. To address this issue, Zheng et al. (2018) uses an ad hoc formulation inspired by force-directed graph drawing:

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which is compatible with an effective learning rate $\lambda \leq 1$. Since f_{ar}^G is bounded within $[-0.5, 0]$, this learning-rate range is stable. For $\zeta < d$, by contrast, $f_{ar}^{MDS} > 0$ exhibits repulsive behavior. Overall, this attraction-repulsion interaction works best when the initialization is already close to the desired output. Continued optimization of a pair causes oscillation around the target distance ($\zeta = d$), where $f_{ar}^G = 0$, giving a distance notion analogous to UMAP's ζ_{-1} . No learning-rate choice reduces this distance to zero, since preserving it is precisely the MDS objective. Consequently, MDS clusters are often fuzzier than those produced by methods such as UMAP and PaCMAP; relevant illustrations appear in Lambert et al. (2022), de Bodt et al. (2025), and Kury et al. (2025). Finally, Lambert et al. (2022) converts the MDS loss into quartet stress, involving four samples, and uses relative distances, defined by dividing a distance by the sum of six quartet distances, thereby enabling global regularization and reducing computation relative to majorization.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) [Hinton and Roweis, 2002] underpins many modern dimensionality-reduction algorithms. It constructs a high-dimensional graph for the dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$ through the following equations:

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_i - x_t\|_2^2 / 2\sigma_i^2)}, \quad (87)$$

where σ_i^2 is selected to match the user-defined *perplexity* value P , defined as

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP, in contrast, constructs its high-dimensional graph through a k-nearest-neighbor (k-NN) procedure governed by

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k nearest neighbors of x_i , and σ_i is a scaling parameter chosen so that $\sum_j p_{i|j} = \log_2(k)$. The graph is then symmetrized through a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses only the k-NN graph for affinities, all set to 1, together with a self-tuning distance measure [Zelnik-Manor and Perona, 2004],

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average Euclidean distance from x_i to its fourth through sixth nearest neighbors. Although the three definitions differ, this parameter plays the same conceptual role as the corresponding σ_i parameters in Eq. (87) and (90).

Regardless of these graph-construction choices, our analysis indicates that the optimization dynamics, and especially the attraction and repulsion shapes, are the primary determinants of the low-dimensional embedding.

I Detailed Results for Varying λ_a and λ_b

This section provides detailed results for the experiments summarized in Fig. 1(h,i). We obtained these results by replacing UMAP’s attraction and repulsion shapes with those of other methods. Figure 18 reproduces the main-text results. Figures 19–22 show, for each method and each choice of λ_a and λ_r , the individual embeddings together with their initialization, a reference embedding obtained by annealing the learning rate λ , and embeddings obtained when either λ_a or λ_r is set to 0. For PaCMAP, which uses mid-near pairs, we additionally include a reference embedding in which mid-near pairs are retained. Figures 23–27 and Figures 28–32 report the corresponding results for FMNIST and the single-cell transcriptomes dataset, respectively.

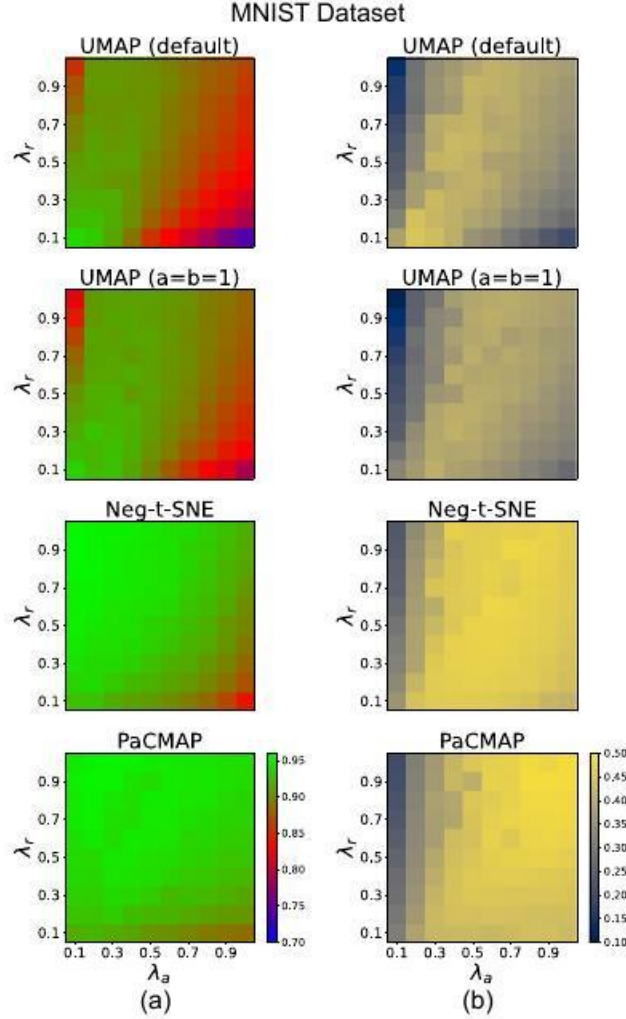


Figure 18: (a) Trustworthiness and (b) silhouette score across methods for the MNIST dataset.

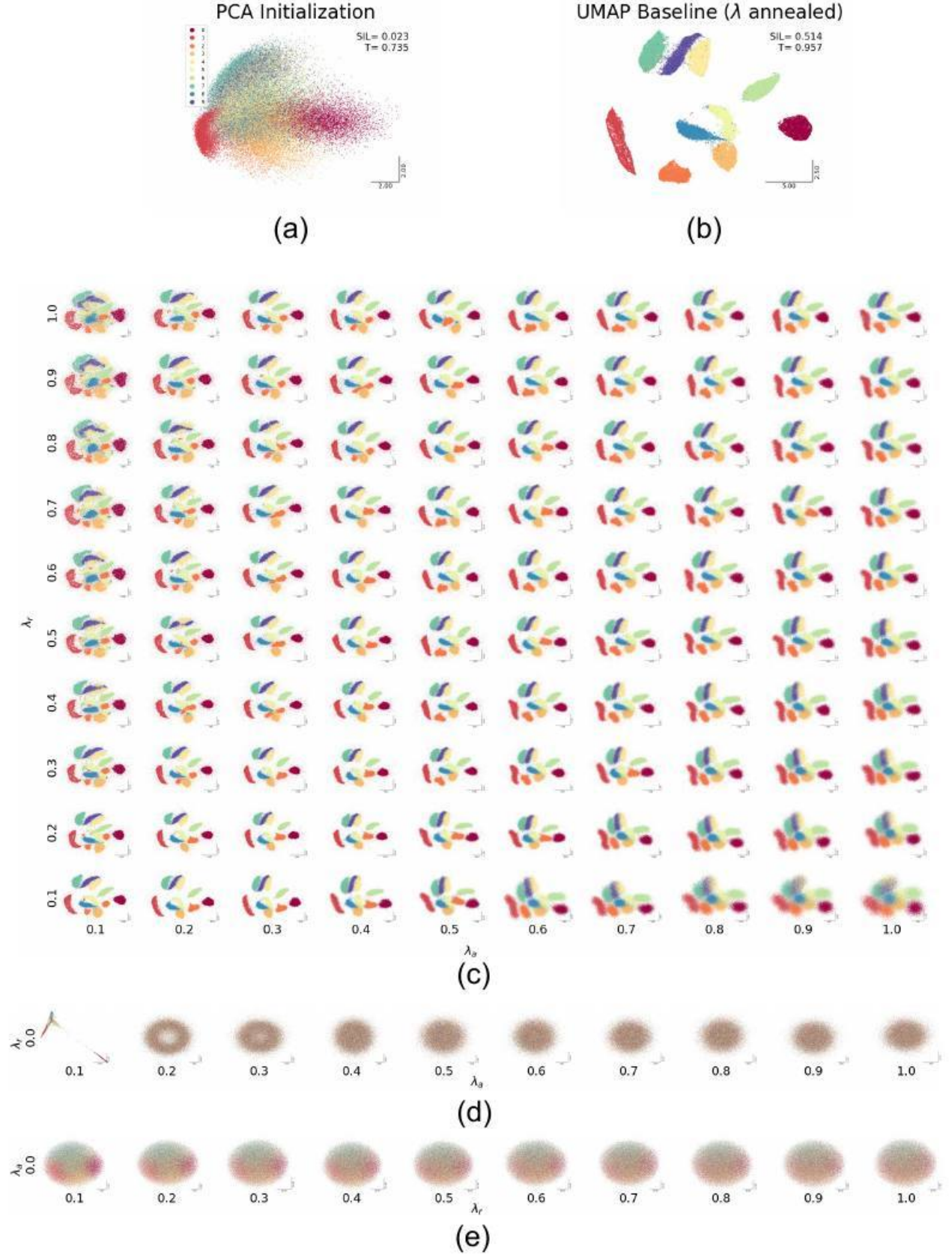


Figure 19: Independent variation of λ_α and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed-learning-rate baseline with λ initialized at 1. (c) Embeddings obtained as λ_α and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_α to 0 (no attraction) shows that repulsion alone cannot produce clusters.

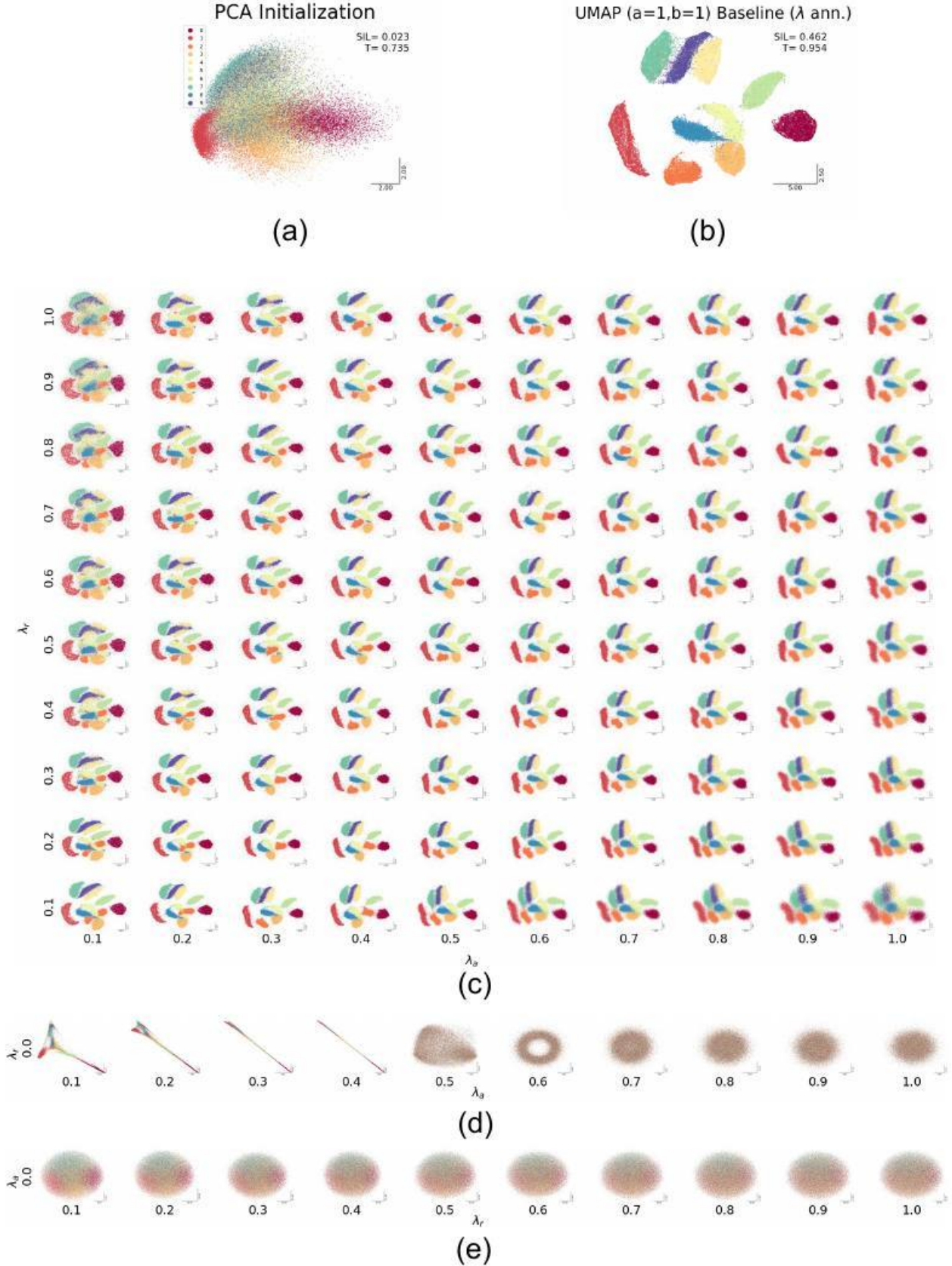


Figure 20: Independent variation of λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Embedding initialization. (b) Annealed-learning-rate baseline with λ initialized at 1. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

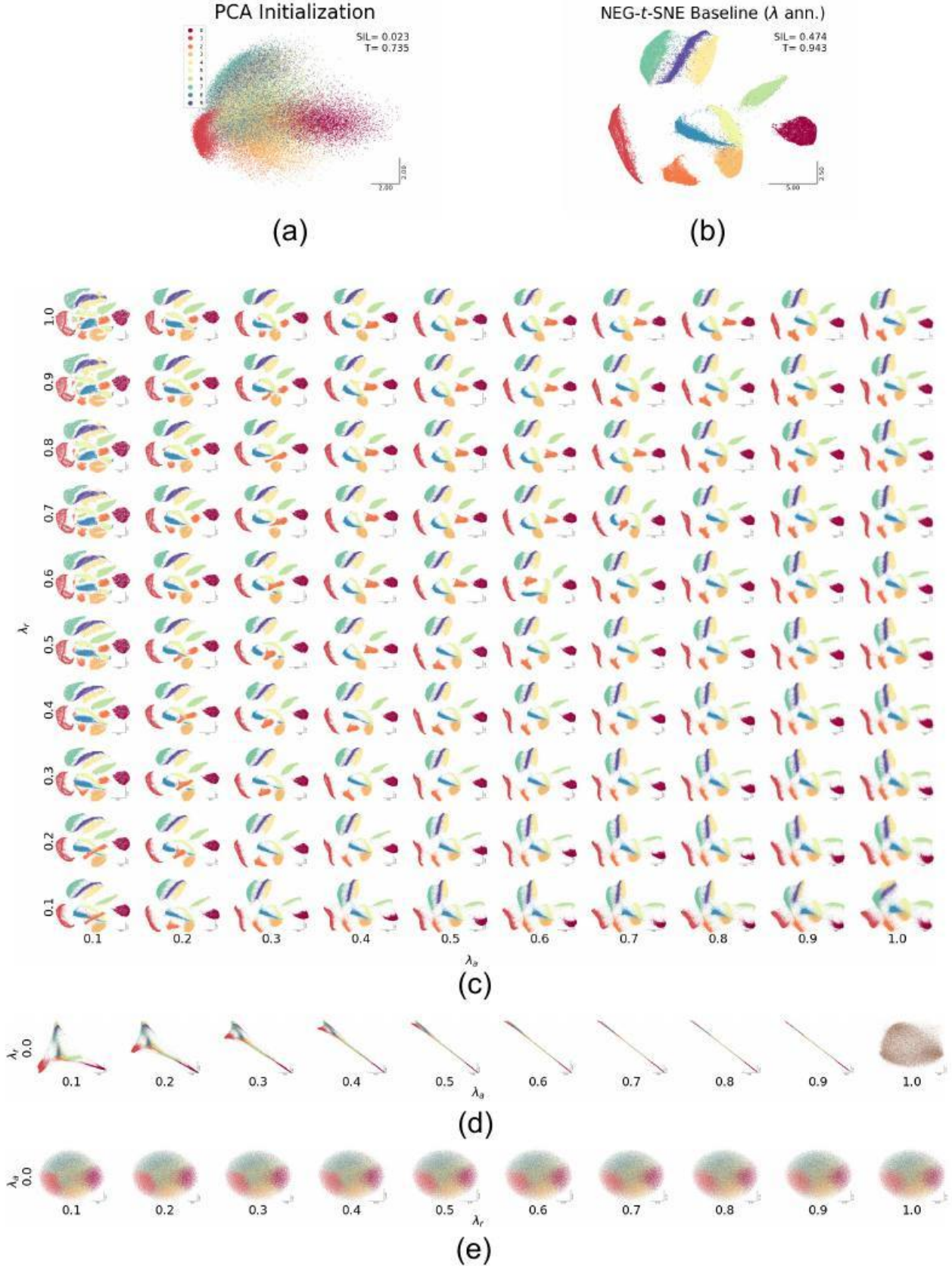


Figure 21: Independent variation of λ_a and λ_r for the MNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed-learning-rate baseline with λ initialized at 1. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

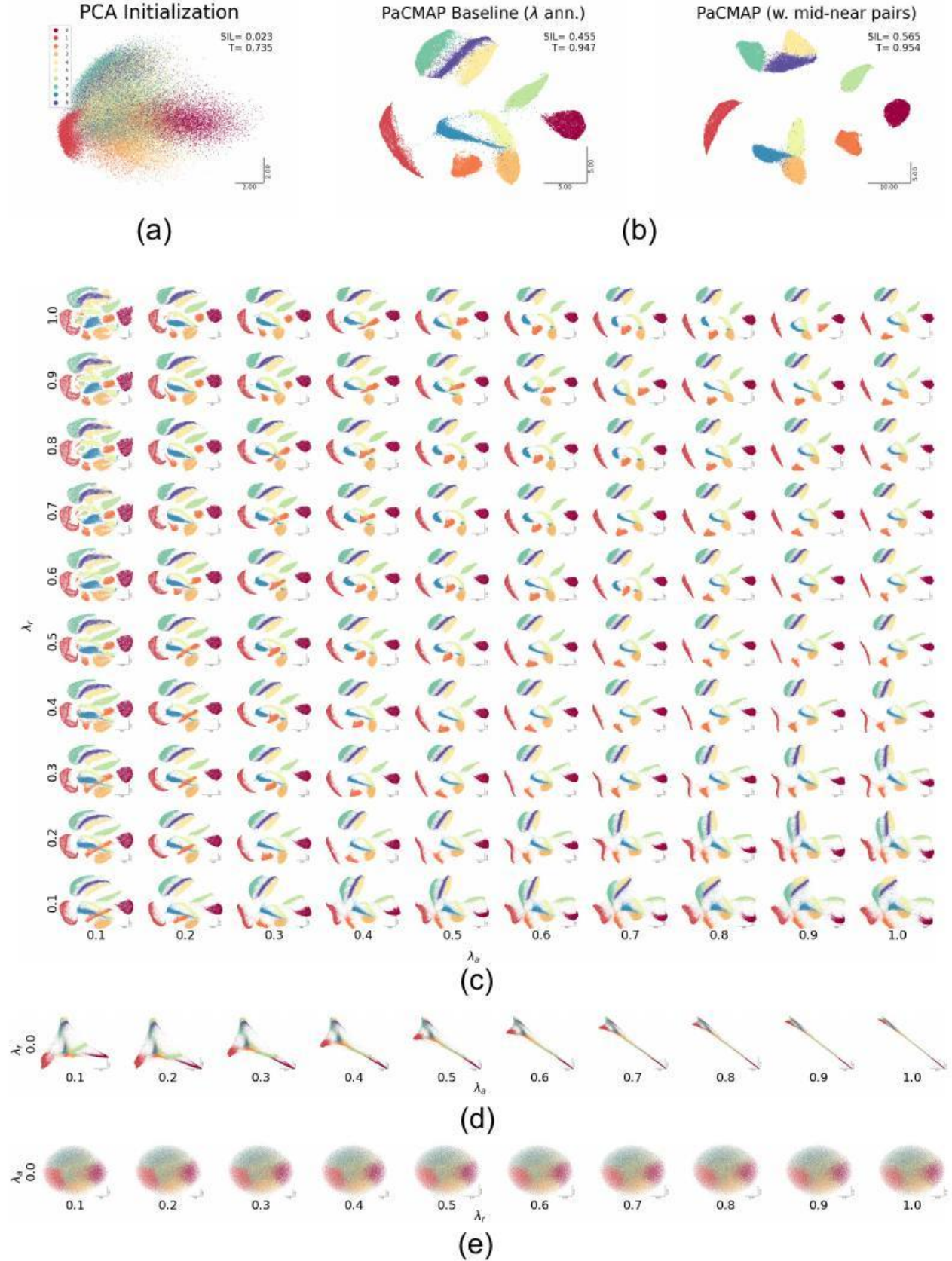


Figure 22: Independent variation of λ_a and λ_r for the MNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Left: annealed-learning-rate baseline with λ initialized at 1, excluding mid-near points to isolate attraction–repulsion interactions; right: baseline with mid-near points included. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

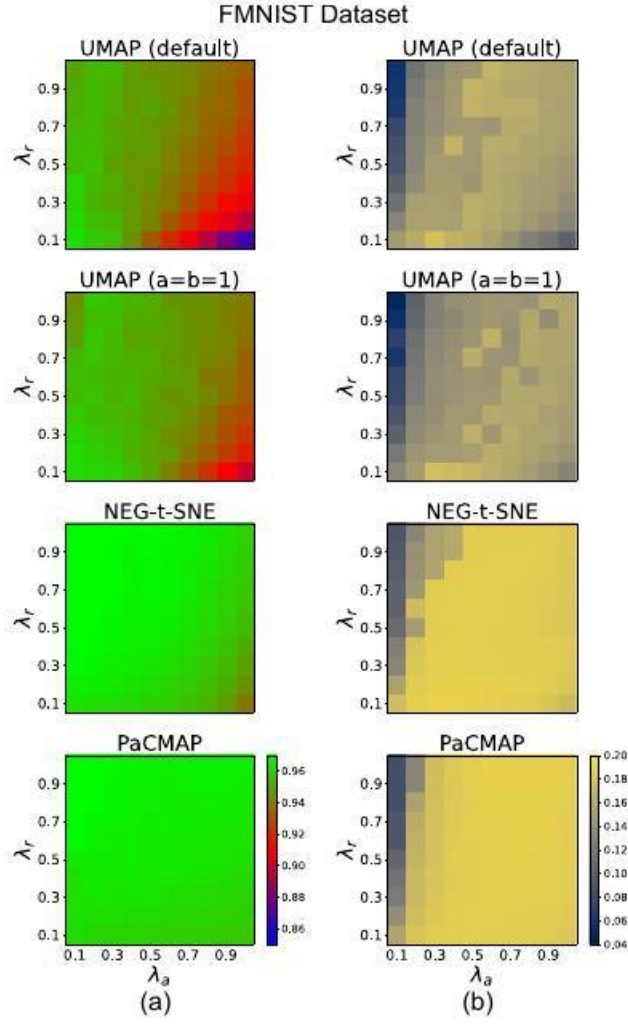


Figure 23: (a) Trustworthiness and (b) silhouette score across methods for the FMNIST dataset.

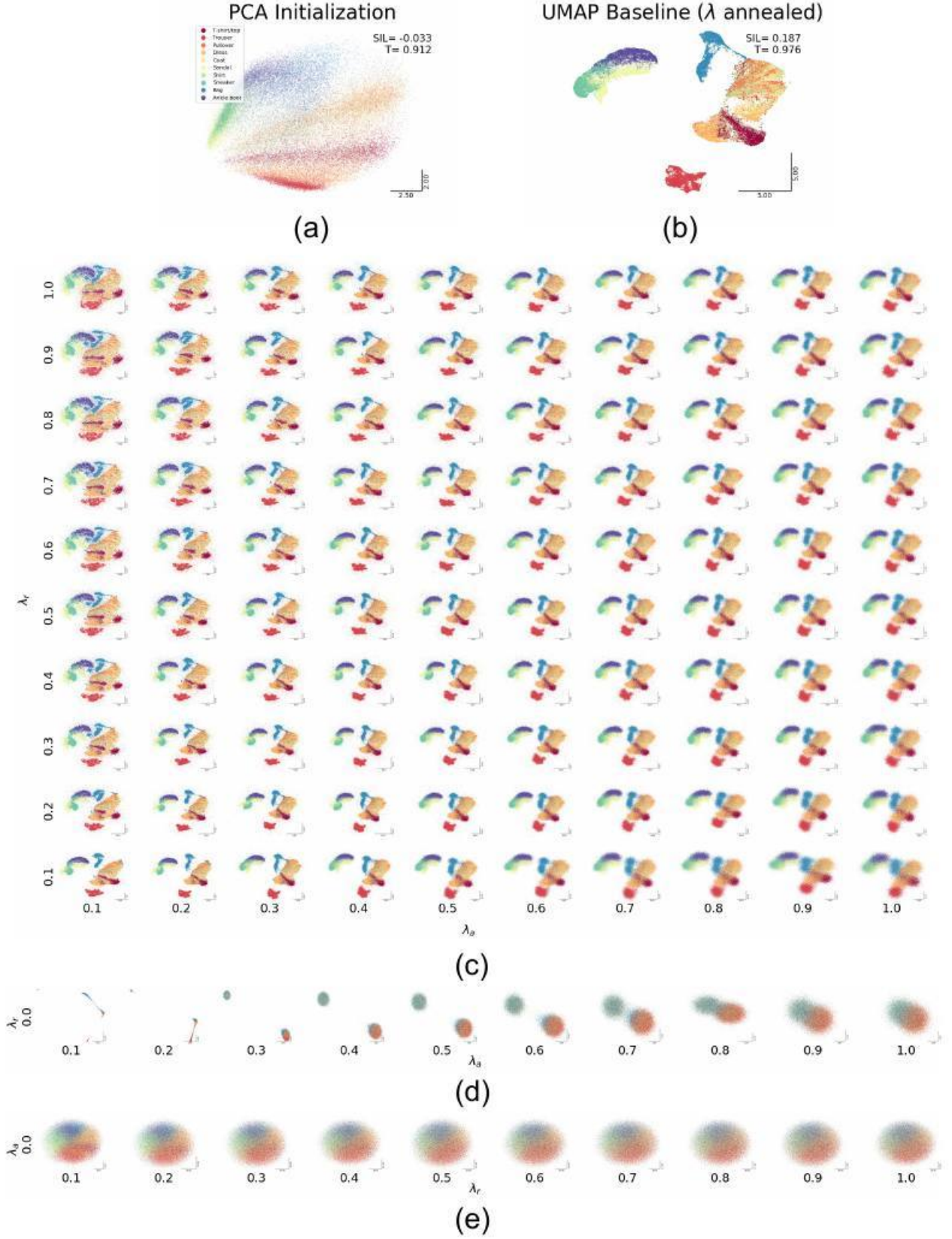


Figure 24: Independent variation of λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed-learning-rate baseline with λ initialized at 1. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

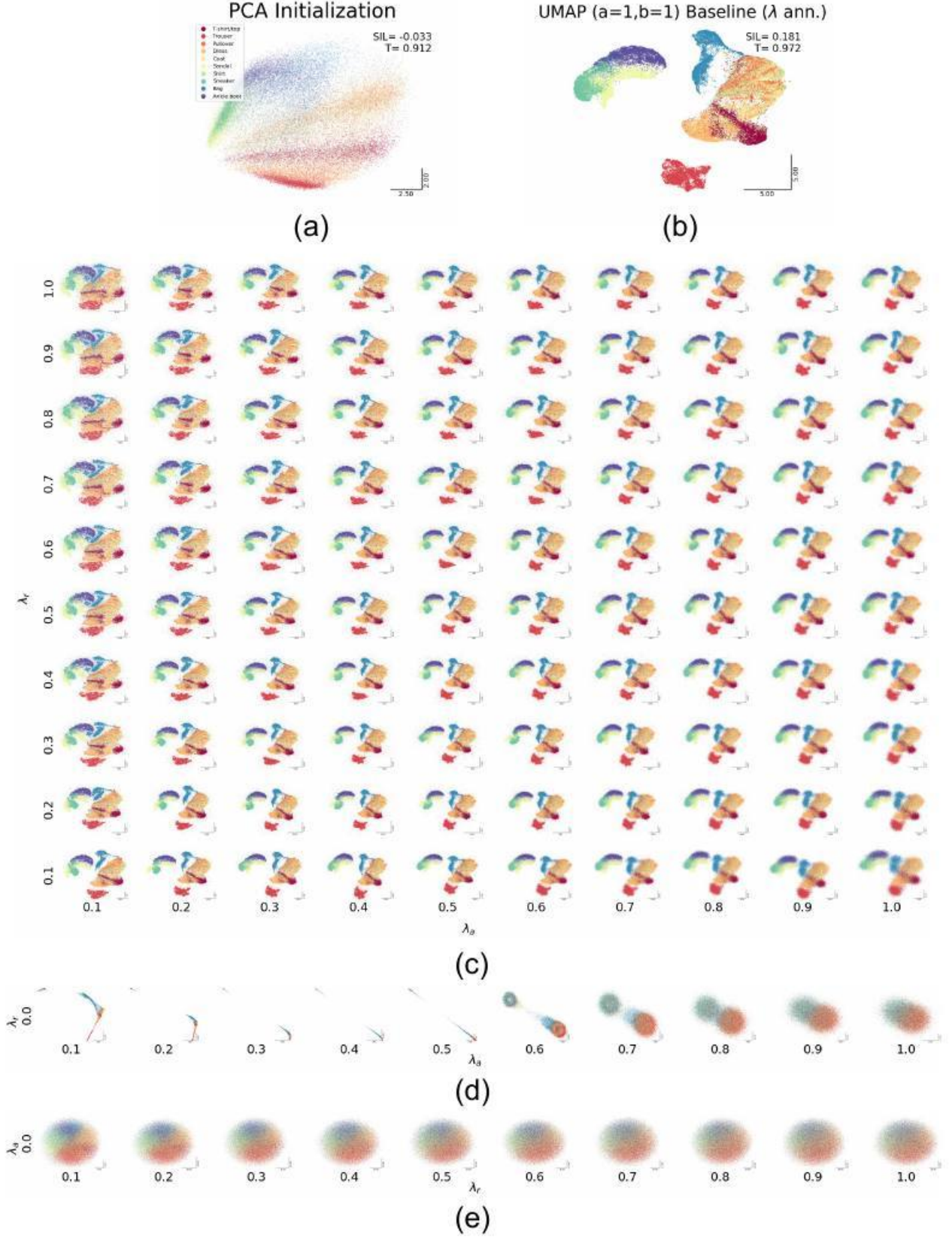


Figure 25: Independent variation of λ_a and λ_r for the FMNIST dataset using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Embedding initialization. (b) Annealed-learning-rate baseline with λ initialized at 1. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

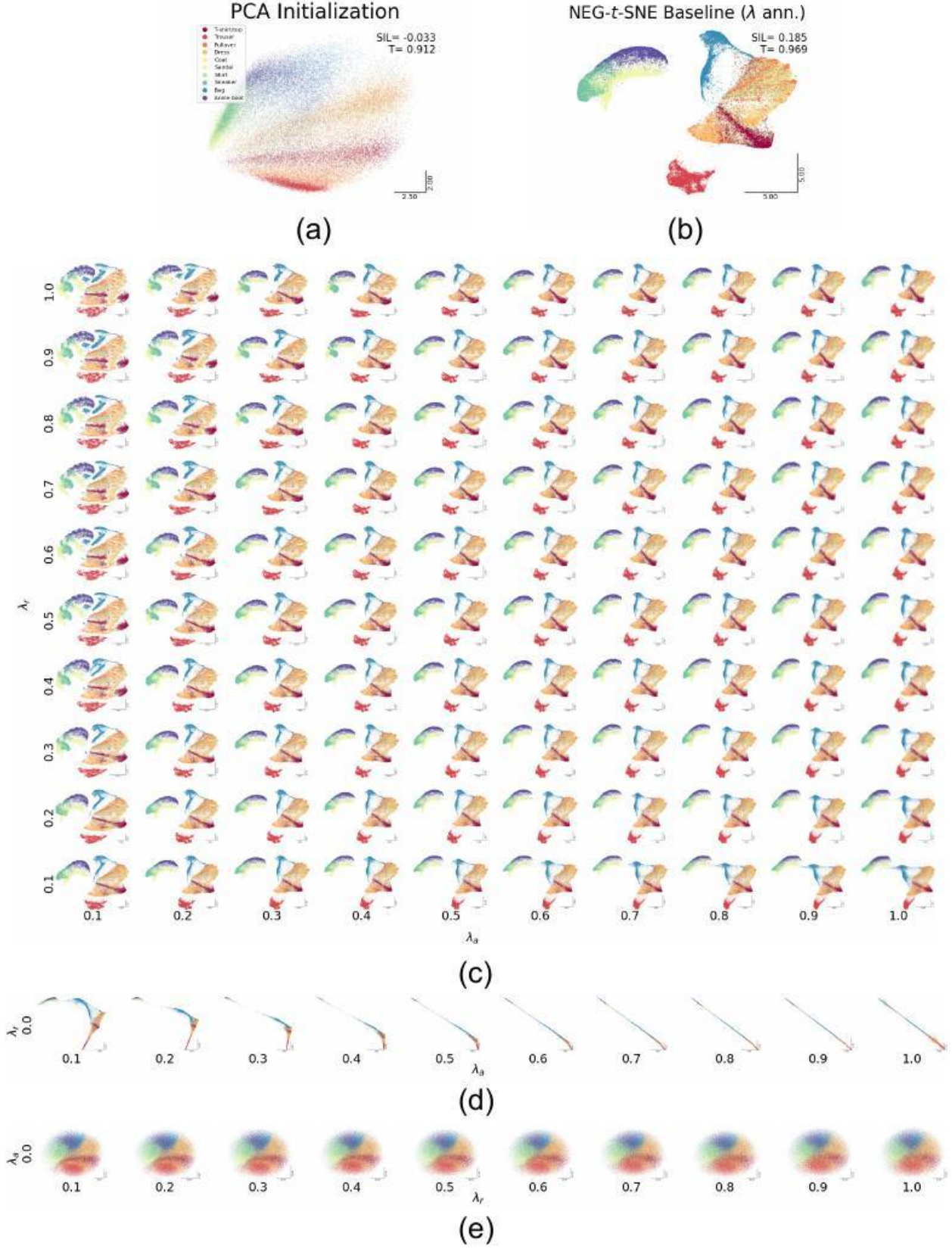


Figure 26: Independent variation of λ_a and λ_r for the FMNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed-learning-rate baseline with λ initialized at 1. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

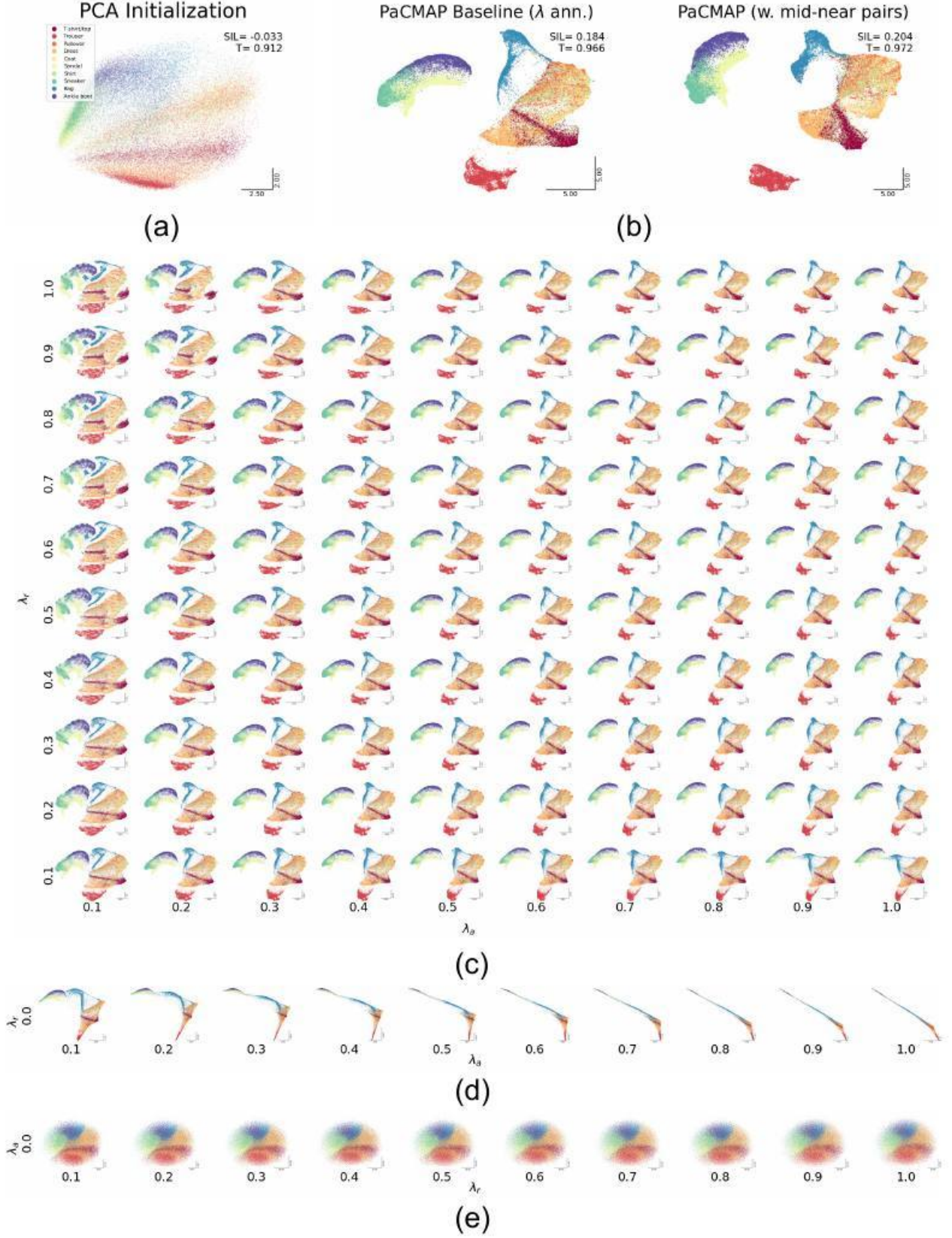


Figure 27: Independent variation of λ_a and λ_r for the FMNIST dataset using PaCMAP’s attraction and repulsion shapes. (a) Embedding initialization. (b) Left: annealed-learning-rate baseline with λ initialized at 1, excluding mid-near points to isolate attraction–repulsion interactions; right: baseline with mid-near points included. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

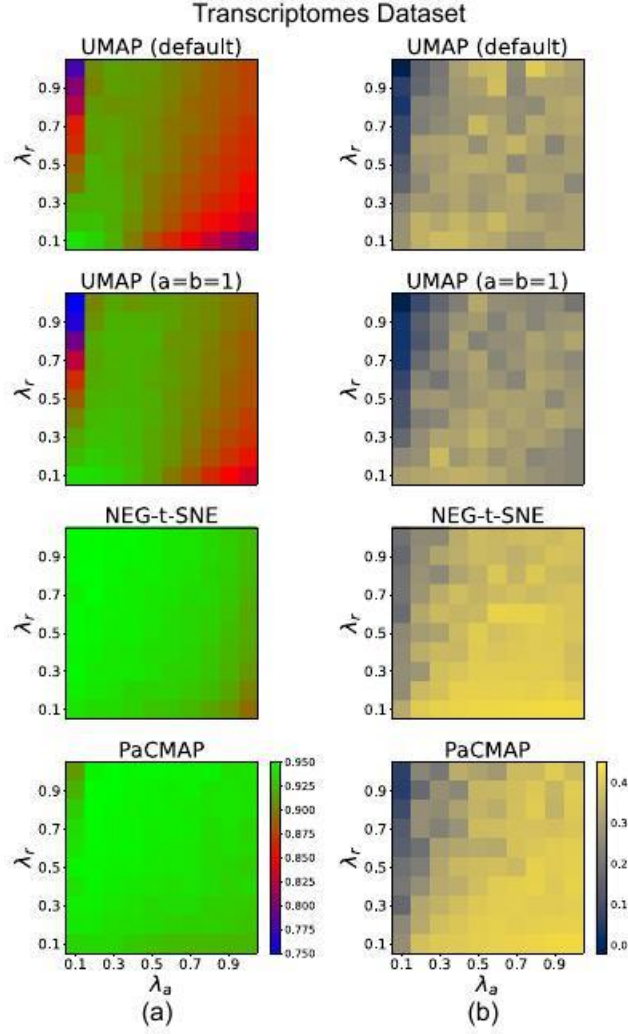


Figure 28: (a) Trustworthiness and (b) silhouette score across methods for the Single-cell transcriptomes dataset.

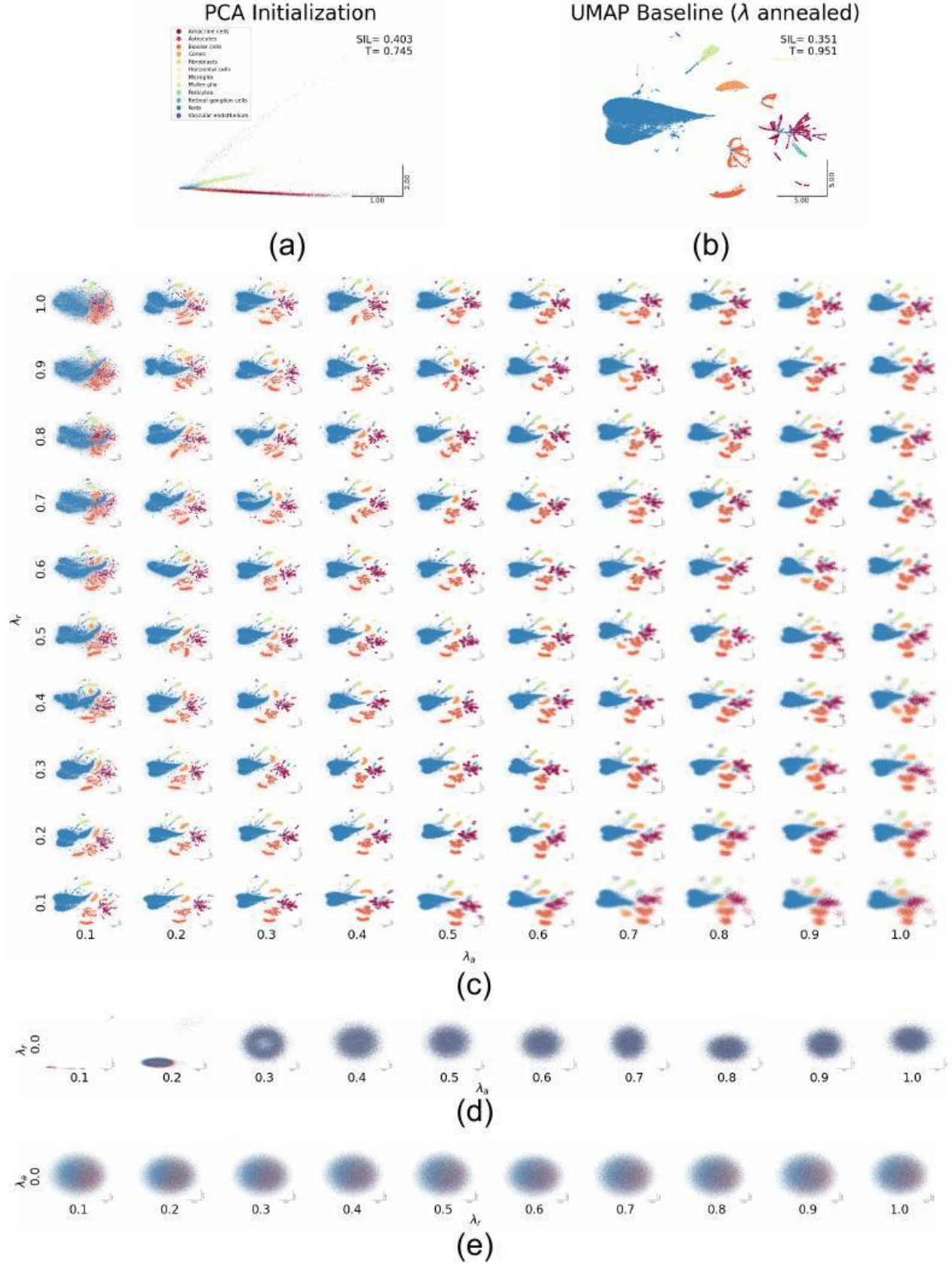


Figure 29: Independent variation of λ_a and λ_r for the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed-learning-rate baseline with λ initialized at 1. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

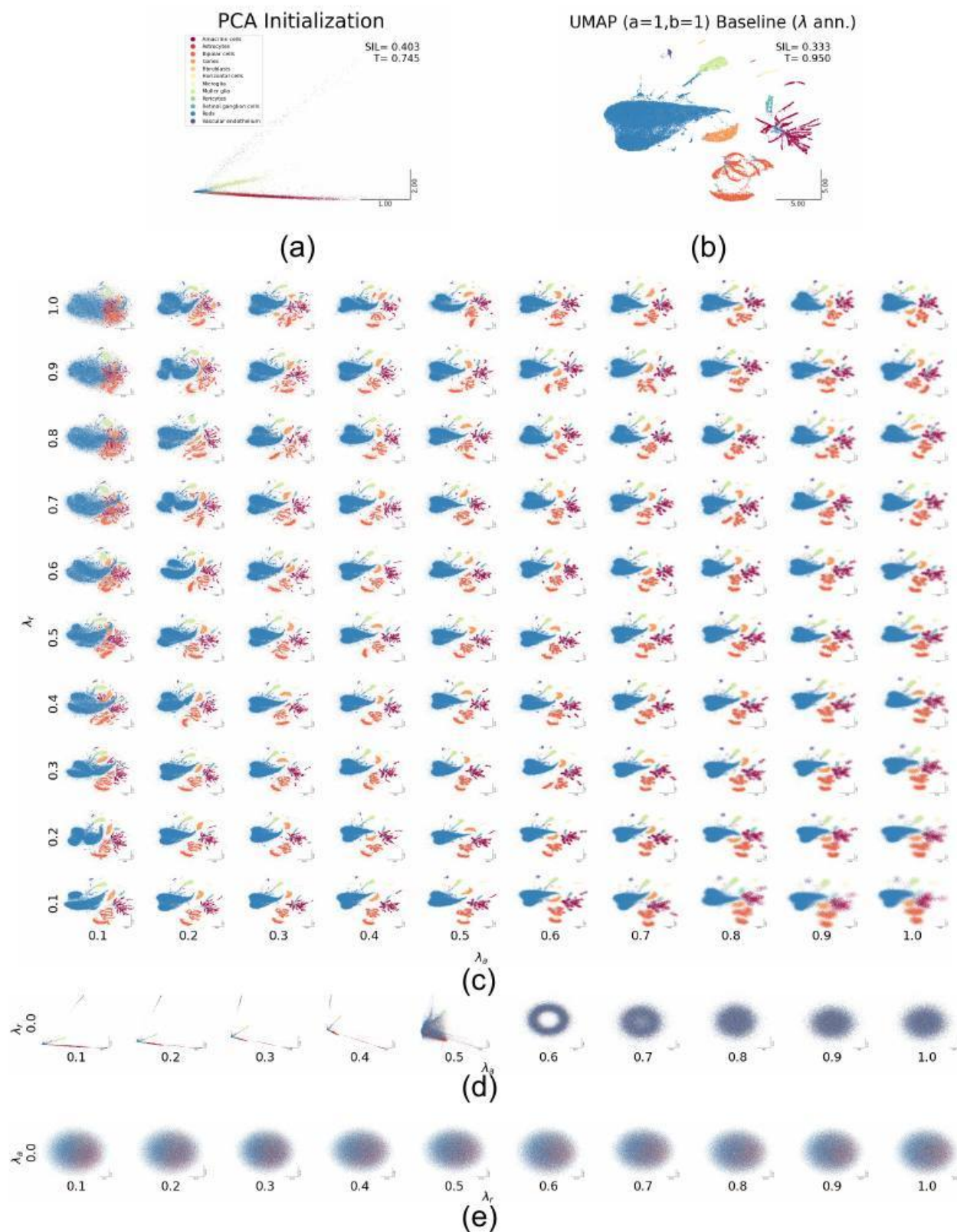


Figure 30: Independent variation of λ_a and λ_r for the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Embedding initialization. (b) Annealed-learning-rate baseline with λ initialized at 1. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

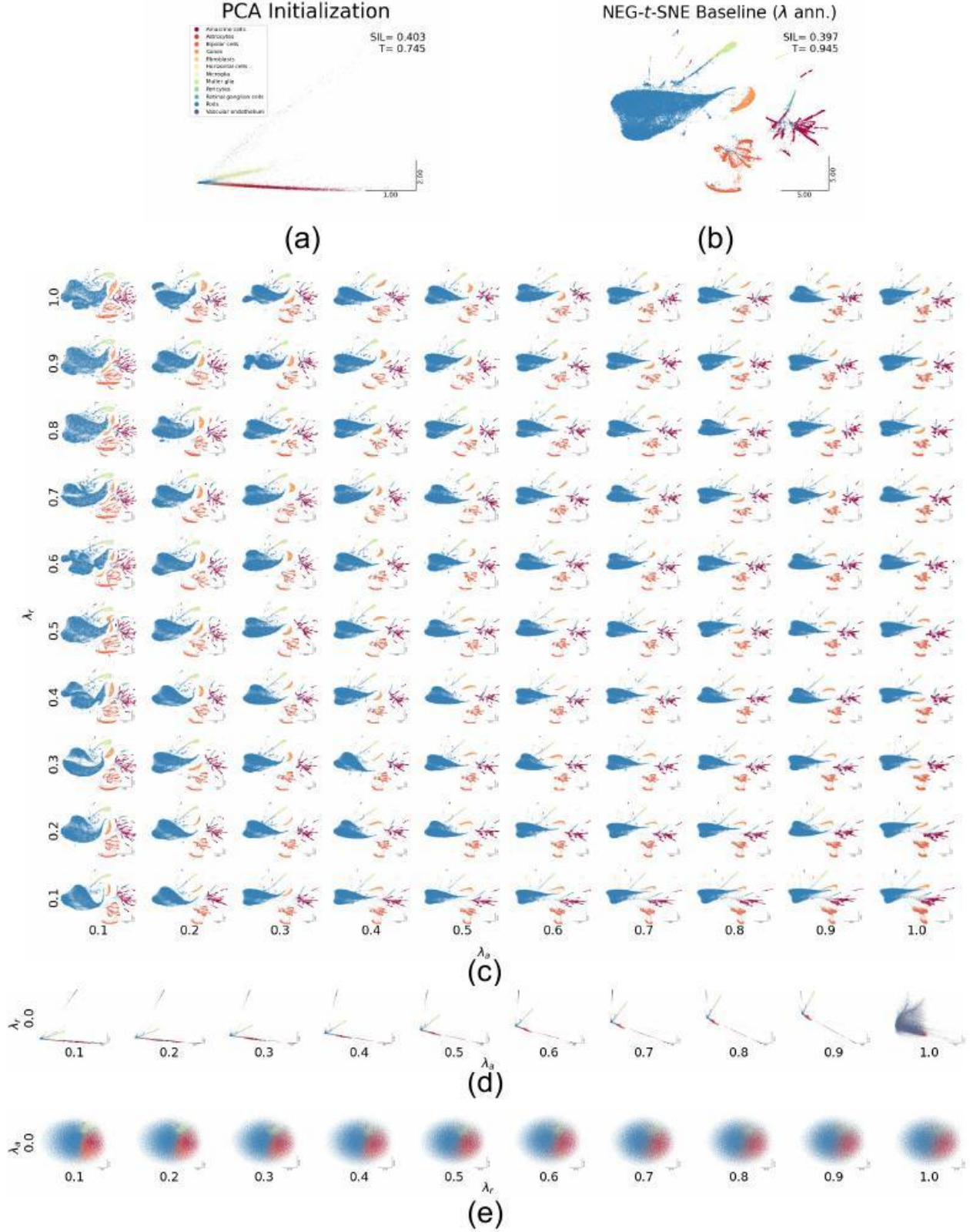


Figure 31: Independent variation of λ_a and λ_r for the single-cell transcriptomes dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Embedding initialization. (b) Annealed-learning-rate baseline with λ initialized at 1. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce distinct clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

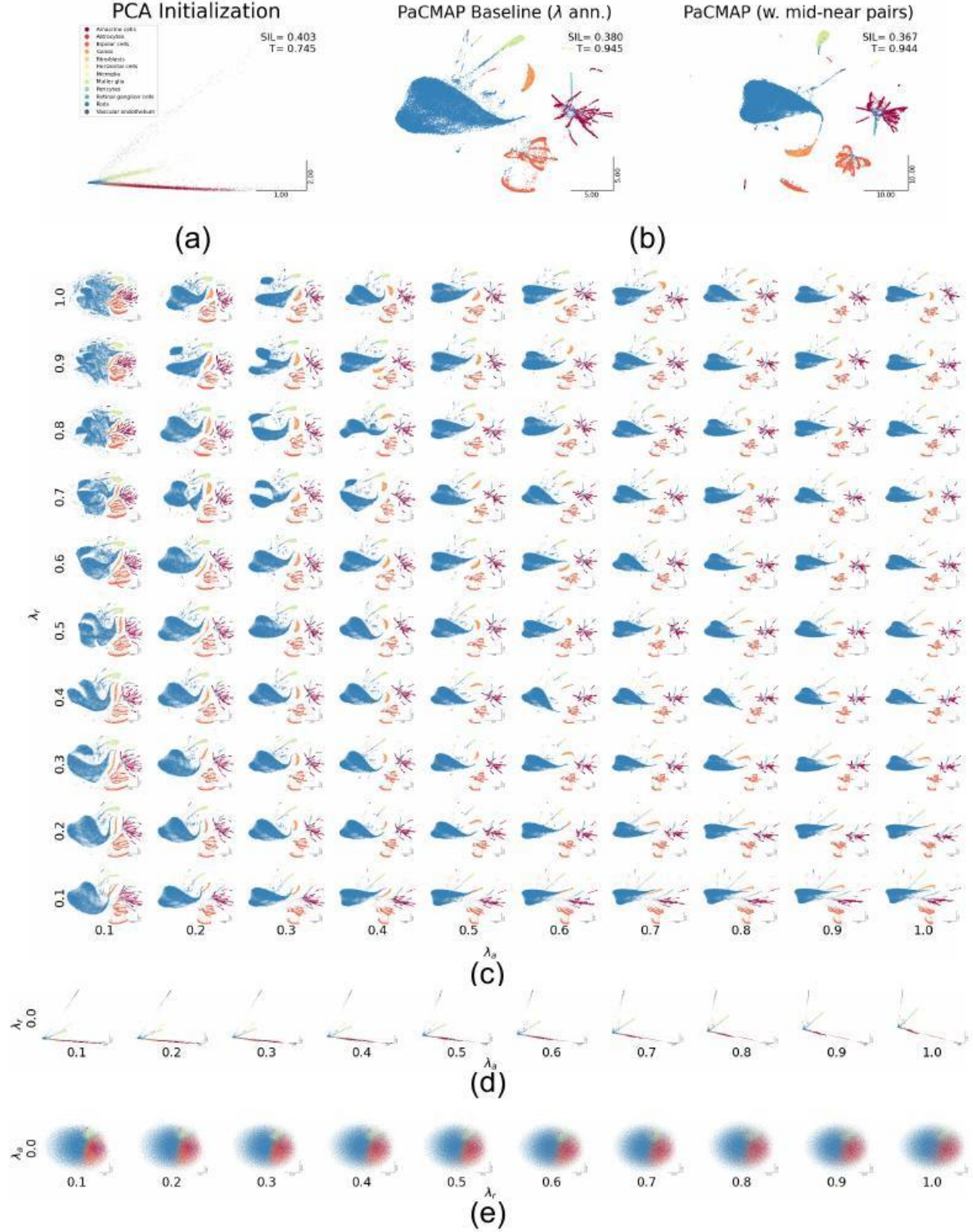


Figure 32: Independent variation of λ_a and λ_r for the single-cell transcriptomes dataset using PaCMAP's attraction and repulsion shapes. (a) Embedding initialization. (b) Left: annealed-learning-rate baseline with λ initialized at 1, excluding mid-near points to isolate attraction–repulsion interactions; right: baseline with mid-near points included. (c) Embeddings obtained as λ_a and λ_r vary without annealing. (d) Setting λ_r to 0 (no repulsion) shows that attraction alone cannot produce clusters. (e) Similarly, setting λ_a to 0 (no attraction) shows that repulsion alone cannot produce clusters.

J Mix and Match Approach

Our analysis treats attraction and repulsion shapes as separable functional components, which suggests that shapes derived from different loss functions and algorithmic principles can be recombined. Figure 33 presents several such examples. In these experiments, we restrict attraction shapes to $[-1, 0]$ so that Proposition 4.1 applies and the updates become less sensitive to learning rate, and we use finite repulsion shapes to mitigate the effect of very large repulsive forces at small distances. We keep the learning rate fixed at $\lambda = 1$ and use the same UMAP optimization procedure throughout, ensuring that observed changes arise from the shapes rather than from learning-rate scheduling or optimizer differences. Overall, UMAP’s attraction shape, scaled as $f_a = 0.5f_a^U$, produces better clusters, reflected in a higher silhouette score, whereas PaCMAP’s attraction shape yields better local structure, reflected in higher trustworthiness. The latter effect primarily follows from PaCMAP’s attraction shape lying within $[-0.5, 0]$, which prevents flips during updates. Constraining the UMAP and NEG- t -SNE attraction shapes to $[-0.5, 0]$ through $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively, improves local structure at the cost of clustering performance (Fig. 34).

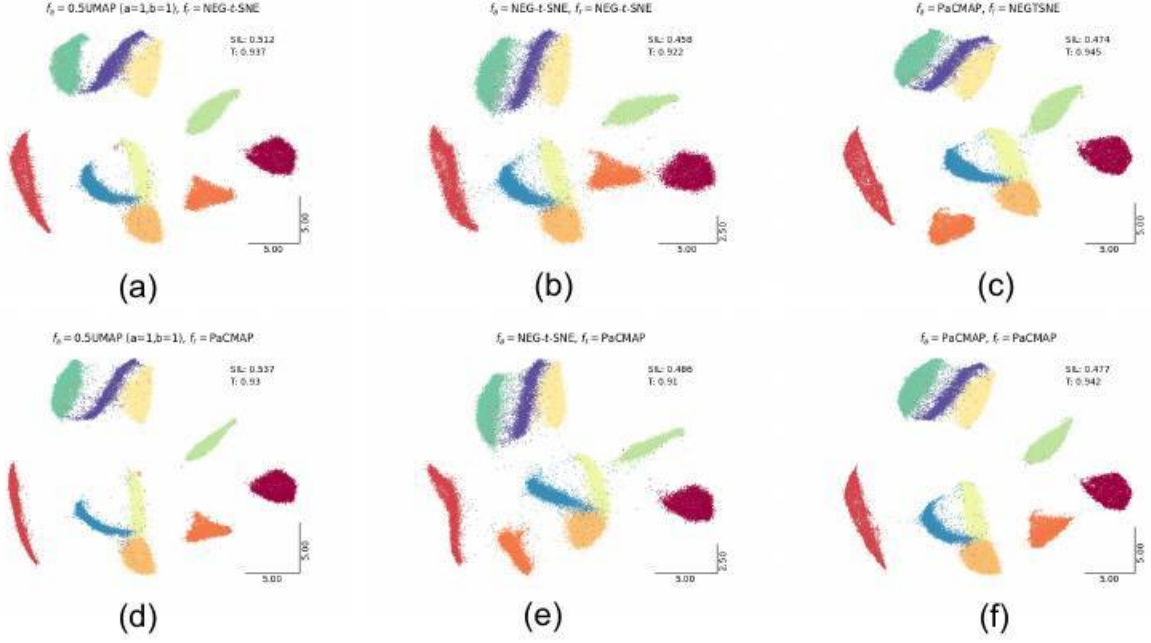


Figure 33: MNIST embeddings produced by mixing attraction and repulsion shapes under a constant learning rate ($\lambda = 1$). Attraction shapes satisfy Proposition 4.1 and repulsion shapes are finite. Top row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes combined with NEG- t -SNE’s repulsion shape. Bottom row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes combined with PaCMAP’s repulsion shape.

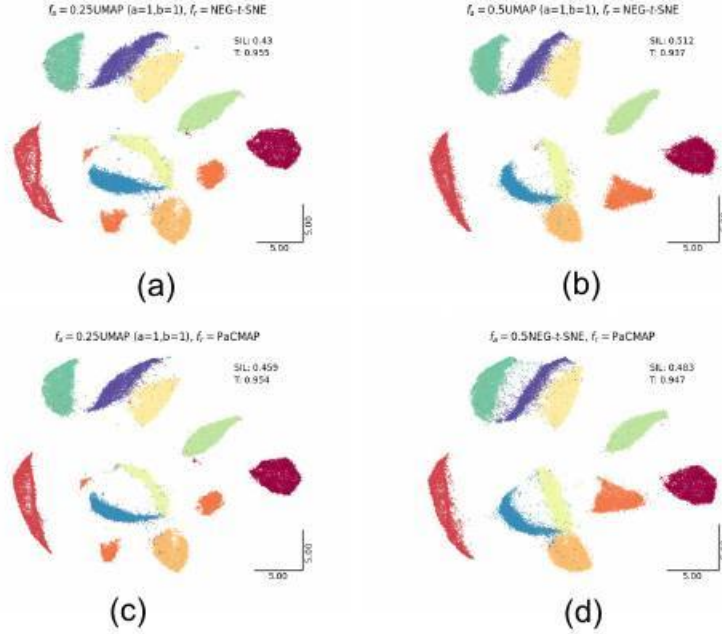


Figure 34: MNIST embeddings produced by mixing attraction and repulsion shapes under a constant learning rate ($\lambda = 1$), with attraction confined to $[-0.5, 0]$ to prevent flips during attractive updates. Top row: UMAP and NEG- t -SNE attraction shapes paired with NEG- t -SNE’s repulsion shape. Bottom row: UMAP and NEG- t -SNE attraction shapes paired with PaCMAP’s repulsion shape.

K Implementation Details

For the analysis, we implemented a custom UMAP algorithm. We used `numba` (Lam et al. 2015) to compute an exact, rather than approximate, nearest-neighbor graph with $k = 15$, and we used `scikit-learn`'s (Pedregosa et al. 2011) PCA implementation for PCA initialization. This implementation produced and quantified the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9, and 10. Unless otherwise stated, we used the same implementation when substituting attraction and repulsion shapes from alternative methods (Figs. 18–34). Trustworthiness and silhouette scores were computed using the corresponding functions in `scikit-learn`.

The mappings in Figs. 2, 3, 9, 10, and 15 are rotated to their corresponding reference embeddings, panel (a) in each figure. We performed this Procrustes alignment by normalizing embeddings to zero mean and unit norm and then using SciPy's (Virtanen et al. 2020) `orthogonal_procrustes` routine to extract rotation and scaling parameters.

For the comparison with Neg- t -SNE in Fig. 11, we used the original contrastive-embedding framework implementation for both UMAP and Neg- t -SNE; implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22 (b) (right), 27 (b) (right), and 32 (b) (right) were generated with the official PaCMAP package; implementation details will be released upon acceptance.

The code used to display the PubMed embeddings in Fig. 7 will be released upon acceptance.